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# Desert Hedgehog mediates stem Leydig cell differentiation through Ptch2/Gli1/Sf1 signaling axis

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## eLife Assessment

This study provides **valuable** contributions to establish canonical Dhh signaling as a primary mediator in the differentiation of Leydig cells and their steroidogenic capacity. Together, the experimental design using their established stem Leydig cell line alongside relevant genetically mutated models, both derived using the relevant Nile tilapia animal system, provided largely **convincing** evidence to support their conclusions. The work could benefit from a more rigorous dissection of current literature on this pathway that might better inform their conclusions. The work will be of broad interest to developmental biologists interested in differentiation of steroidogenic or hormone producing cells.

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## Abstract

Desert Hedgehog (Dhh) mutations cause Leydig cell dysfunction, yet the mechanisms governing Leydig lineage commitment through Dhh-mediated receptor selectivity, transcriptional effector specificity, and steroidogenic coupling remain elusive. In this study, using CRISPR/Cas9-mediated gene knockout and stem Leydig cells (SLCs) transplantation, we identified a critical Dhh/Patched 2 (Ptch2)/Glioma-associated oncogene homolog 1 (Gli1)/steroidogenic factor 1 (Sf1) signaling axis essential for SLC differentiation in Nile tilapia (*Oreochromis niloticus*). Dhh deficiency resulted in defective adult Leydig cells and androgen insufficiency. Rescue experiments involving 11-ketotestosterone administration and a Dhh agonist treatment, combined with SLCs transplantation, demonstrated that Dhh regulates SLC differentiation, not survival. *In vitro* knockout of *ptch1* and *ptch2* in SLCs revealed that Ptch2 likely acts as the functional receptor for Dhh. This was further supported by *in vivo* genetic rescue experiments, where *ptch2* mutation did not impair testicular development, yet completely rescued the testicular defects in *dhh* mutants—consistent with Ptch2 acting as an inhibitory receptor whose loss alleviates Dhh pathway suppression. Luciferase assays in Gli-knockout SLCs demonstrated that Gli1 acts as the primary transcriptional effector, and transactivates *sf1* expression. Additionally, functional transplantation assays confirmed that Sf1 is indispensable for SLCs differentiation, as Sf1-overexpressing SLCs rescued differentiation, whereas *sf1*-mutant SLCs failed. Overall, our work delineates the Dhh-Ptch2-Gli1-Sf1 axis and provides fundamental insights into the endocrine regulation of Leydig cell lineage development.

## Highlights

- Dhh regulates the differentiation of SLCs rather than their survival

- Ptch2 acts as the functional receptor for Dhh signaling in SLCs
- Gli1 is the principal transcriptional activator for Dhh signaling in SLCs
- Sf1 is the critical downstream effector of Gli1 in SLCs differentiation

## Introduction

Leydig cells are the primary source of androgens, producing approximately 95% of circulating testosterone to sustain spermatogenesis and male fertility [1, 2]. Leydig cells originate from stem Leydig cells (SLCs), which differentiate into steroidogenic adult Leydig cells (ALCs) under paracrine regulation [3, 4]. Steroidogenic factor 1 (Sf1), an orphan nuclear receptor transcription factor, serves as a master regulator of this process by activating steroidogenic enzymes (e.g., cytochrome P450, family 11, subfamily C, polypeptide 1 [Cyp11c1], 3 $\beta$ -hydroxysteroid dehydrogenase [3 $\beta$ -HSD]) to drive ALC maturation [5]. In mice, mutations of *Sf1* exhibit arrested testicular development and abolished steroidogenesis [6–8], while human *SF1* mutations are associated with 46,XY disorders of sexual development (DSD), such as Swyer syndrome [9]. Similarly, in Nile tilapia (*Oreochromis niloticus*), our previous work reveals that *sf1* mutation disrupts testicular morphogenesis and downregulates the expression of key steroidogenic enzymes such as *cyp11c1* [10]. Despite its established central role, the upstream signaling pathways that regulate Sf1 to commit the Leydig lineage remain poorly defined.

Hedgehog (Hh) signaling pathway is an evolutionarily conserved developmental regulator [11]. Canonical Hh signaling involves the binding of ligands to Patched (Ptch) receptors, which relieves the suppression on Smoothened (Smo) and ultimately activates Glioma-associated oncogene homolog (Gli) transcription factors to regulate target genes [12–14]. In vertebrates, there are three Hh ligands (i.e. Sonic Hh, Indian Hh and Desert Hh [Dhh]), two Ptch receptors (Ptch1 and Ptch2) and three Gli homologs (Gli1–3) [15–18]. A large number of studies have well demonstrated that Dhh signaling pathway is critical in gonadal development [14, 19]. Dhh, secreted by Sertoli cells, is critical for Leydig cell commitment in mammals [20, 21]. In mice, mutations of *Dhh* lead to Leydig cell dysfunction, testosterone insufficiency, and infertility [22], while human *DHH* mutations are associated with 46,XY DSD [23]. Strikingly, these phenotypes mirror those caused by *Sf1* deficiencies, suggesting a potential regulatory nexus between Dhh and Sf1 signaling. However, the molecular circuitry connecting Dhh signaling to Sf1 activation remains elusive.

Studies demonstrate functional divergence between Ptch1 and Ptch2. For instance, *Ptch1* knockout causes embryonic lethality due to constitutive Hh activation, whereas *Ptch2* deficiency yields no overt defects [24, 25]. During testicular development, Ptch1 displays high expression at early stage critical for gonadal primordium formation, while Ptch2 shows testis-enriched dynamic regulation [26, 27]. However, the distinct roles of Ptch1 and Ptch2 in Leydig lineage commitment remain unknown to date. Among Gli transcription factors, Gli1 acts as a primary activator, while Gli2 and Gli3 exhibit dual regulatory functions [28, 29]. In murine testes, co-expression of Gli1 and Gli2 in Leydig cells illustrates functional redundancy—single knockouts show normal development, but combined pharmacological inhibition (GANT61) abolishes Leydig cells and spermatogenesis [30]. These findings underscore the complexity of Gli-mediated transcription in the testis and highlight the need to delineate the specific roles of individual Gli in SLC differentiation.

Nile tilapia (*Oreochromis niloticus*) is not only an important farm fish in global aquaculture but also a distinguished animal model for investigating gene function and endocrine regulation. This model organism possesses multiple advantageous attributes, including an XX/XY sex-determination system, early sexual maturity (approximately 3 months after hatching for males and 6 months after hatching for females), a short spawning cycle of approximately 14 days, year-round reproductive capacity under laboratory conditions, well-characterized sex-linked genetic markers, and a body size amenable to physiological sampling [31, 32]. Furthermore, a stem Leydig cell line (TSL) has been established from the testis of a 3-month-old Nile tilapia [33]. TSL expresses platelet-derived growth factor receptor  $\alpha$  (*pdgfra*), *nestin*, and chicken ovalbumin upstream promoter transcription factor II (*coup-fla*), which are usually considered as SLC-related markers in several other species [2, 34, 35]. Notably, this cell line exhibits the capacity to differentiate into 11-ketotestosterone (11-KT)-producing Leydig cells both *in vitro* and *in vivo* [33]. When cultured in

a defined induction medium, TSL cells differentiate into a steroidogenic phenotype, expressing key steroidogenic genes including *star1*, *star2*, and *cyp11c1*, and producing 11-KT; upon transplantation into recipient testes, TSL cells successfully colonize the interstitial compartment, activate the expression of steroidogenic genes, and restore 11-KT production. In this study, by combining CRISPR/Cas9-mediated gene knockout with TSL transplantation technology, we unveil a Dhh-Ptch2-Gli1-Sf1 axis that is essential for Leydig cell lineage differentiation. Our findings provide fundamental insights into the endocrine regulation of Leydig cell development.

## Results

### Mutation of *dhh* disrupts testicular organization and androgen synthesis

The Nile tilapia Dhh open reading frame (GenBank ID: 100707022, 1377 bp) was cloned and validated. The encoded 458-amino acid protein shares over 56% identity with vertebrate homologs (Fig. S1A) and clusters phylogenetically with mammalian DHH orthologs (Fig. S1B). Transcriptomic analyses conducted on various tissues and testes across different developmental stages revealed that *dhh* expression peaked in the testes, particularly at 90 days after hatching (dah) (Fig. S1C, D).

CRISPR/Cas9 targeting of *dhh* exon 1 from Nile tilapia generated homozygous mutants (*dhh*<sup>-/-</sup>) harboring a 13-bp frameshift deletion (Fig. S2A–D), resulting in premature termination and loss of the conserved Hedge domain (Fig. S2E). *dhh* mRNA levels in *dhh*<sup>-/-</sup> fish were significantly reduced compared to WT (Fig. S2F). At 90 dah, *dhh*<sup>-/-</sup> testes exhibited marked atrophy (Fig. 1A, B) and disorganized histology, with sparse spermatogonia, rare spermatids, and collapsed interstitial architecture (Fig. 1C–D', K). Immunofluorescence confirmed significantly reduced germ cell (Vasa<sup>+</sup>), meiotic cell (Sycp3<sup>+</sup>), and Leydig cell (Cyp11c1<sup>+</sup>) populations (Fig. 1E–J', L–N). Serum 11-KT levels in *dhh*<sup>-/-</sup> fish were significantly lower than WT (Fig. 1O), indicating impaired androgen synthesis.

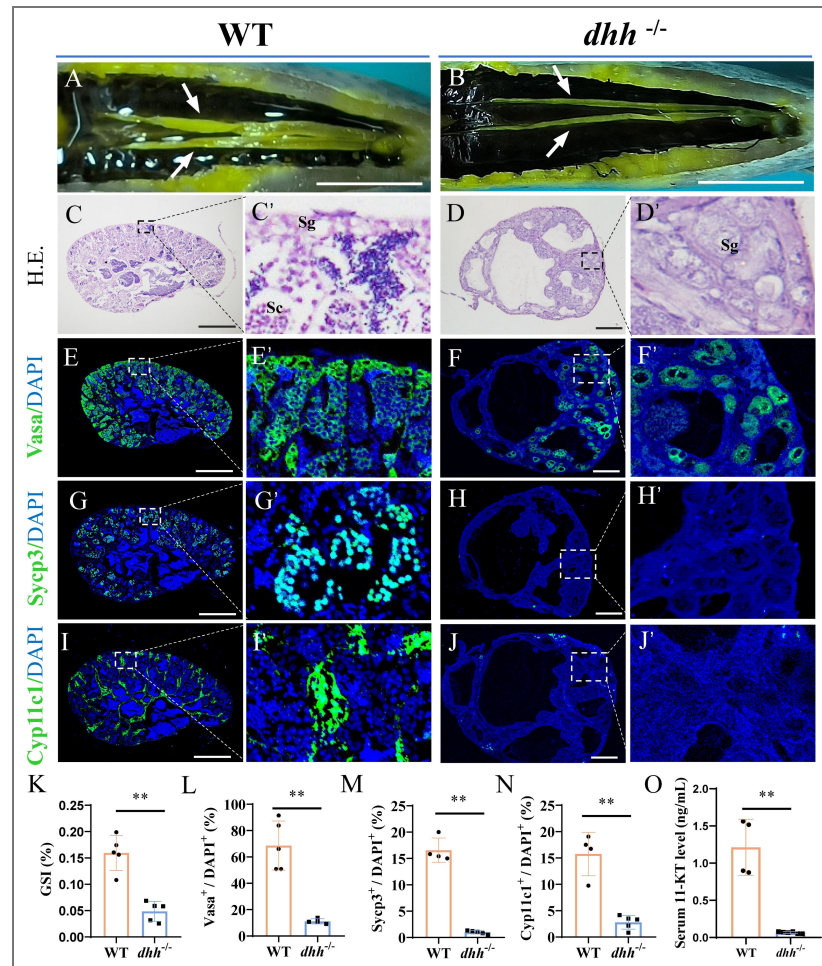
### The differentiation of SLCs cannot be rescued by 11-KT, but by SAG

Given the established role of 11-KT in teleost testicular development and spermatogenesis [36], we first asked whether exogenous 11-KT could ameliorate the testicular defects in *dhh*<sup>-/-</sup> mutants. Compared to untreated mutants, *dhh*<sup>-/-</sup> individuals exposed to 11-KT exhibited a more organized testicular architecture, with significantly increased populations of Vasa<sup>+</sup> and Sycp3<sup>+</sup>, as evidenced by both H&E staining and immunofluorescence (Fig. 2A–C', E–G', I–K', R, S). This improvement was further supported by a recovery in GSI (Fig. 2Q). However, the population of Cyp11c1-positive Leydig cells remained profoundly depleted in *dhh*<sup>-/-</sup> +11-KT testes (Fig. 2M–O', T). Taken together, these data indicate that 11-KT treatment partially restored the development of germ cells but not Leydig cell lineage, implying that the Leydig cell defect is androgen-independent.

We next investigated whether direct activation of the Dhh pathway could rescue the *dhh*<sup>-/-</sup> phenotype. Treatment of *dhh*<sup>-/-</sup> mutants with SAG, a Hedgehog agonist, significantly rescued testicular development. This was marked by a significant recovery of germ cells (Vasa<sup>+</sup>), spermatocytes (Sycp3<sup>+</sup>), and critically, Leydig cells (Cyp11c1<sup>+</sup>), alongside a restoration of testicular histology (Fig. 2D–D', H–H', L–L', P–P', Q–T). These results demonstrate that *dhh* mutation does not disrupt survival of endogenous Leydig cell lineage, but directly impairs their differentiation, leading to secondary impairment in androgen production and germ cell development.

### Mutation of *dhh* blocks SLCs differentiation *in vivo*

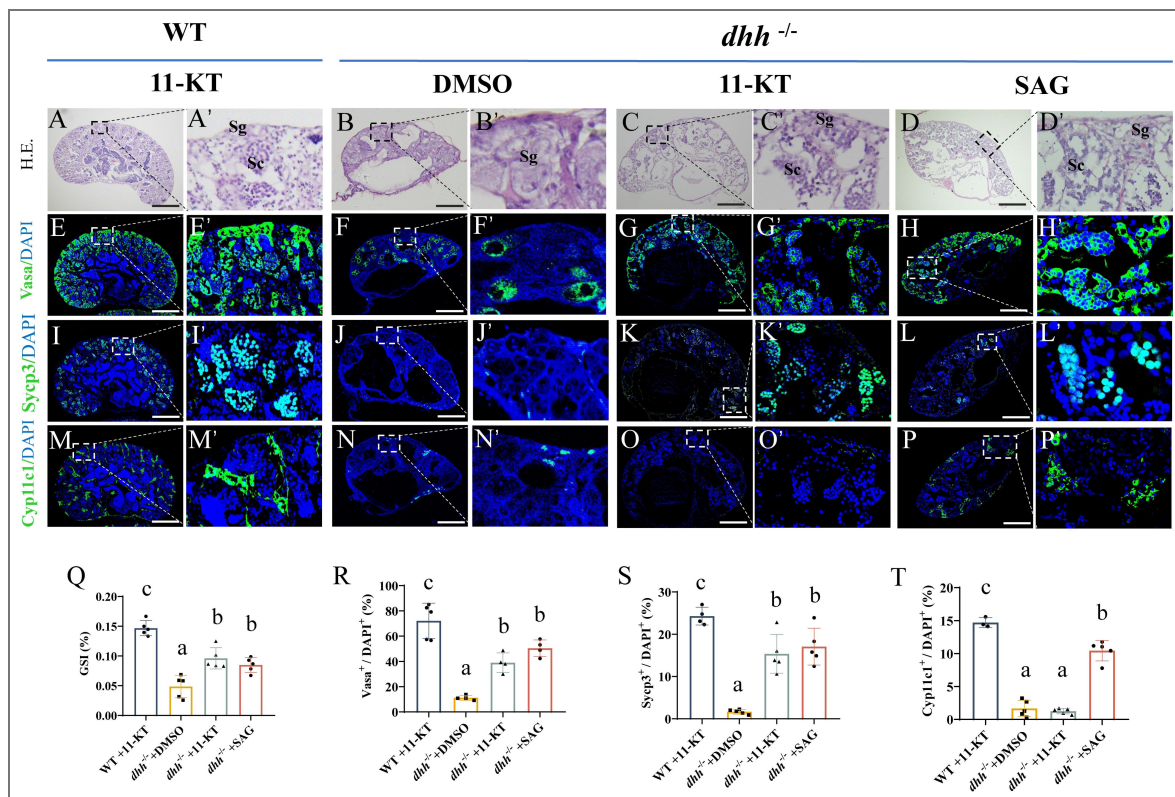
To dissect Leydig cell lineage impairment in *dhh*<sup>-/-</sup> testes, we transplanted the TSL labeled with PKH26 (a fluorescent red hydrophobic membrane dye that enables tracking of transplanted cells) into WT and *dhh*<sup>-/-</sup> testes (Fig. 3A). In WT recipients, TSL cells colonized the interstitium and differentiated into Cyp11c1 positive steroidogenic cells (Fig. 3B1–B3). In *dhh*<sup>-/-</sup> testes, TSL engrafted equivalently but failed to express Cyp11c1 (Fig. 3C1–C3), indicating a differentiation-specific block. Rescue experiments showed that the TSL overexpressing tilapia Dhh (TSL-*OnDhh*)



**Fig. 1. Mutation of *dhh* resulted in testicular developmental disorders.**

(A-B) Morphological analysis of the testes from WT (*dhh*<sup>+/+</sup>) and *dhh*<sup>-/-</sup> XY fish at 90 dah. Arrows indicate the location of the testes. (C-D') Histological analysis of testicular sections from WT and *dhh*<sup>-/-</sup> XY fish at 90 dah. Sg, spermatogonia; Sc, spermatocyte. (E-J') Representative immunofluorescence images showing the expression of germ cell marker Vasa (E-F'), meiosis cell marker Sycp3 (G-H') and Leydig cell marker Cyp11c1 (I-J'). (K) Gonadosomatic index (GSI) of the testes from WT and *dhh*<sup>-/-</sup> XY fish at 90 dah (*n*=5 fish/genotype). (L-N) Quantification of the percentage of cells positive for Vasa (E-F'), Sycp3 (G-H') and Cyp11c1 (I-J') among all DAPI-positive cells in the WT and *dhh*<sup>-/-</sup> testes (*n*=5 fish/genotype). (O) Serum 11-KT level of WT and *dhh*<sup>-/-</sup> XY fish at 90 dah (*n*=6 fish/genotype). Values were presented as mean ± SD. Differences were determined by two-tailed independent Student's t-test. \*\*, *P* < 0.01. Scale bars, (A-B), 1 cm; (C-J), 50 μm.





**Fig. 2. Rescue of testicular development in *dhh*<sup>-/-</sup> XY Nile tilapia by 11-KT and SAG.**

*dhh*<sup>-/-</sup> XY Nile tilapia at 30 dah were treated either with 11-KT via water immersion (with water changed every two days) or with 10 mg/kg SAG via intraperitoneal injection (with supplemental injection every seven days). WT controls received 11-KT, and *dhh*<sup>-/-</sup> controls received an equivalent volume of the DMSO. Then, at 90 dah, morphological and histological experiments were conducted. (A-D') Histological analysis of testicular sections from 90 dah XY fish subjected to different treatments as indicated. Sg, spermatogonia; Sc, spermatocyte. Scale bars, 50  $\mu$ m. (E-P') Representative immunofluorescence images showing the expression of Vasa (E-H'), Sycp3 (I-L') and Cyp11c1 (M-P'). Scale bars, 50  $\mu$ m. (Q) GSI of testes from the different treatment groups at 90 dah ( $n=5$  fish per group). (R-T) Quantification of the percentage of cells positive for Vasa (E-H'), Sycp3 (I-L') and Cyp11c1 (M-P') among all DAPI-positive cells in the testes ( $n=3-5$  fish per group). Values were presented as mean  $\pm$  SD. Different letters above the error bar indicate statistical differences at  $P < 0.05$  as determined by one-way ANOVA followed by Tukey test.

or treated with SAG (TSL+SAG) restored Cyp11c1 expression in *dhh*<sup>-/-</sup> testes (Fig. 3D1–E3, F) and a significant increase in endogenous 11-KT levels (Fig. 3G). These results suggest that SLC differentiation is inhibited in *dhh*<sup>-/-</sup> XY tilapia testes. In addition, the survival and engraftment of PKH26-labeled TSL cells were unaffected in the *dhh*<sup>-/-</sup> environment (Fig. 3C3). Collectively, these results demonstrate that Dhh signaling is specifically required for the differentiation of SLCs into steroidogenic cells, but is dispensable for their survival or initial recruitment into the testicular niche.

## Ptch2 is the functional receptor for Dhh signaling in SLCs

RT-PCR results showed that both *ptch1* and *ptch2* were expressed in TSL (Fig. S3). Fluorescence *in situ* hybridization (FISH) and IF revealed co-expression of *ptch1* and *ptch2* in Cyp11c1 positive Leydig cells (Fig. 4A). To identify the Dhh-driving receptor selectivity in SLCs, we established *ptch1*<sup>-/-</sup> and *ptch2*<sup>-/-</sup> TSL cell lines (Fig. S4A, B) and evaluated pathway activity using an 8×GLI luciferase reporter system (Fig. 4B). In WT TSL cells, the 8×GLI reporter demonstrated substantial luciferase activity that was markedly enhanced by *OnDhh* overexpression (Fig. 4C), confirming Dhh responsiveness in this system. Intriguingly, *ptch1*<sup>-/-</sup> cells maintained Dhh-induced luciferase activation comparable to WT controls, whereas *ptch2*<sup>-/-</sup> cells completely lost this responsiveness (Fig. 4D). This differential receptor requirement implies that Ptch2 likely acts as the functional receptor for transducing Dhh signals in TSL cells.

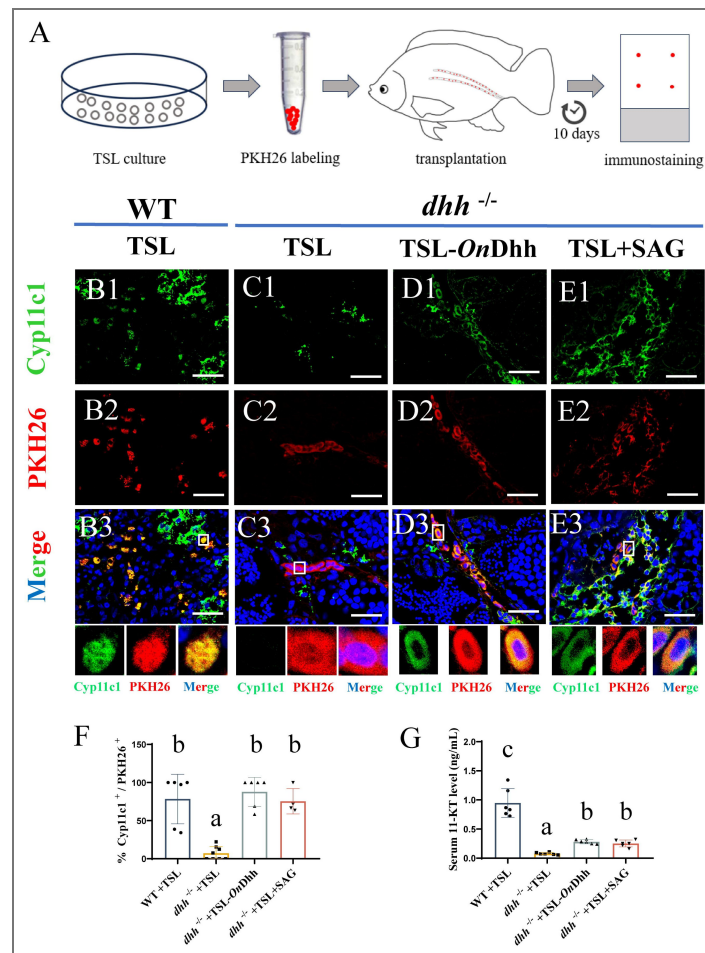
To investigate the function of Ptch2 *in vivo*, we generated *ptch2*<sup>-/-</sup> and *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> mutant Nile tilapias (Fig. S5A–E). Compared to WT fish at 90 dah, *ptch2*<sup>-/-</sup> testes exhibited no significant differences in testicular histology (Fig. 4E–F'), GSI (Fig. 4Q), germ cell (Vasa<sup>+</sup>, Sycp3<sup>+</sup>) and Leydig cell (Cyp11c1<sup>+</sup>) populations (Fig. 4H–I', K–L', N–O', R–T), and serum 11-KT levels (Fig. 4U). Strikingly, the loss of *ptch2* in the *dhh*<sup>-/-</sup> background (*dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup>) fully rescued the testicular defects, resulting in normal histology (Fig. 4G–G'), Leydig cell differentiation (Fig. 4P–P'), and serum 11-KT levels (Fig. 4U). This genetic rescue is consistent with Ptch2 functioning as the inhibitory receptor for Dhh; its removal alleviates the suppression on the pathway, thereby bypassing the requirement for the Dhh ligand.

## Gli1 is the principal transcriptional effector of Dhh signaling in SLCs

In the Hh pathway, Gli transcription factors (Gli1/2/3) mediate diverse developmental processes (26). RT-PCR results showed that *gli1*, *gli2*, and *gli3* were all expressed in TSL (Fig. S3). FISH and IF revealed co-expression of *gli1*, *gli2*, and *gli3* in Cyp11c1 positive Leydig cells (Fig. 5A). To identify Dhh-specific effectors, we generated *gli1*<sup>-/-</sup>, *gli2*<sup>-/-</sup>, and *gli3*<sup>-/-</sup> TSL lines (Fig. S4C–E) and assessed pathway activity using the 8×GLI luciferase reporter. While TSL-*gli2*<sup>-/-</sup> and TSL-*gli3*<sup>-/-</sup> cell lines maintained Dhh-induced luciferase activation comparable to WT controls, the TSL-*gli1*<sup>-/-</sup> cell line completely lost this responsiveness (Fig. 5B), implying that Gli1 is the principal transcriptional effector for Dhh signaling transduction in SLCs.

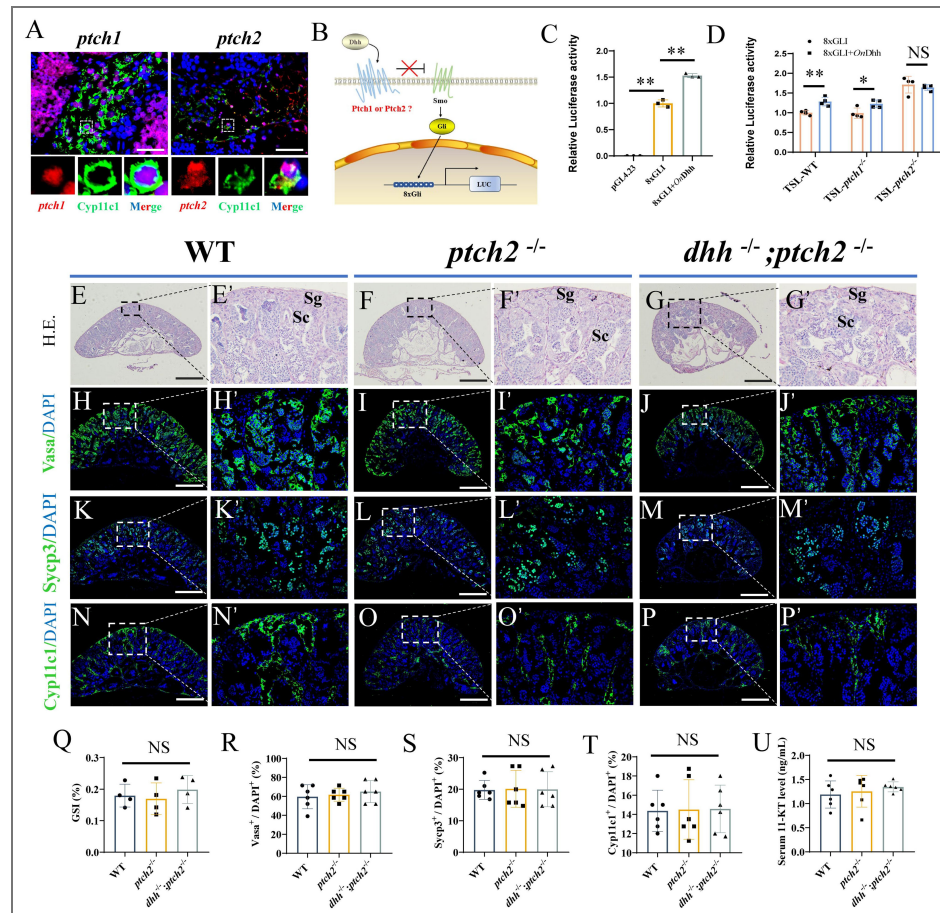
## Sf1 is the critical downstream effector of Gli1 in SLCs differentiation

To identify key targets of the Dhh-Gli1 axis, we performed transcriptomic profiling of TSL cells under conditions of pathway activation: Dhh overexpression (TSL-*OnDhh*), Gli1 overexpression (TSL-*OnGli1*), and SAG treatment (TSL+SAG). Comparative RNA-seq analysis identified a core set of 33 genes consistently upregulated across all three conditions (Fig. 5C, S6A). Among these, *sf1* emerged as a prominently upregulated candidate. Functional annotation of its promoter region identified two conserved Gli1-binding motifs (Fig. 5D). A dual-luciferase reporter assay confirmed that Gli1 potently transactivates the *sf1* promoter (Fig. 5E). The specificity of this interaction was further demonstrated by a cold probe competition assay: a wild-type



**Fig. 3. Dhh is required for SLCs differentiation *in vivo*.**

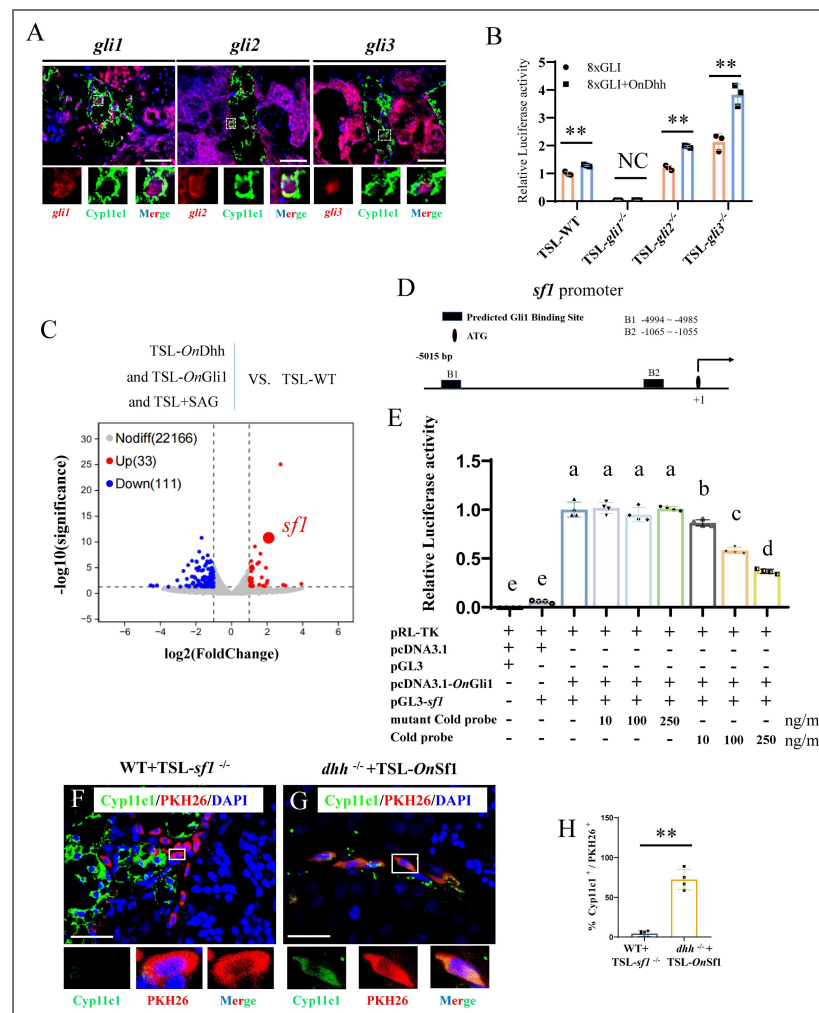
TSL, Dhh-overexpressing TSL (TSL-OnDhh), or 0.5  $\mu$ M SAG-treated TSL (TSL+SAG) cells were labeled with PKH26 and transplanted into the testes of 90 dah WT or *dhh*<sup>-/-</sup> recipient fish. Analyses were performed 10 days post-transplantation. (A) Schematic diagram of the experimental design for SLCs transplantation and analysis. (B1-E3) Representative immunofluorescence images of testicular sections from recipient fish, showing the localization of transplanted PKH26-labeled SLCs (red) and the expression of Cyp11c1 (green). Nuclei are stained with DAPI (blue). Scale bars, 4  $\mu$ m. (F) Quantification of the percentage of Cyp11c1-positive cells among the transplanted PKH26-positive SLCs for each treatment group ( $n=5$  fish per group). (G) Serum 11-KT level in recipient fish following SLCs transplantation ( $n=5$  fish per group). Values were presented as mean  $\pm$  SD. Different letters above the error bar indicate statistical differences at  $P < 0.05$  as determined by one-way ANOVA followed by Tukey test.



**Fig. 4. Ptch2 mediates Dhh signaling in SLCs.**

(A) Co-localization of Cyp11c1 (green, by immunofluorescence) and *ptch1* or *ptch2* mRNA (red, by RNA-FISH) in adult (90 dah) testis sections. Dashed lines outline representative Leydig cells. Scale bars, 4  $\mu$ m. (B) Schematic illustration of the experimental setup for the luciferase reporter assays shown in panels C and D (C) Luciferase activity in TSL cells co-transfected with a Gli-responsive reporter (8xGLI) and overexpression of tilapia Dhh (*OnDhh*). A plasmid lacking Gli-binding sites (pGL4.23) served as a negative control ( $n=3$ ). (D) Luciferase activity in TSL-WT, TSL-*ptch1*<sup>-/-</sup> and TSL-*ptch2*<sup>-/-</sup> cells transfected with the 8xGLI reporter with or without *OnDhh* ( $n=4$ ). (E-G') Histological analysis of testis sections from the indicated genotypes at 90 dah. Sg, spermatogonia; Sc, spermatocyte. Scale bars, 50  $\mu$ m. (H-P') Immunofluorescence analysis of Vasa (H-J'), Sycp3 (K-M') and Cyp11c1 (N-P') in testis sections from the indicated genotypes. Scale bars, 50  $\mu$ m. (Q) GSI of the testes from the indicated genotypes at 90 dah ( $n=4$  fish per group). (R-T) Quantification of the percentage of Vasa (H-J'), Sycp3 (K-M') and Cyp11c1 (N-P') positive cells among DAPI-positive cells ( $n=6$  fish per group). (U) Serum 11-KT levels in fish of the indicated genotypes at 90 dah ( $n=6$  fish per group). Statistical significance was determined by one-way ANOVA followed by Tukey's test (C, Q-U, different letters above the error bar indicate statistical differences at  $P < 0.05$ ) or Student's *t*-test (D) (\*,  $P < 0.05$ ; \*\*,  $P < 0.01$ ; NS, no significant difference).





**Fig. 5. Gli1 transactivates *sf1* to drive SLC differentiation**

(A) Co-localization of Cyp11c1 (green) and *gli1*, *gli2*, or *gli3* mRNA (red) in adult (90 dah) testis sections by immunofluorescence and RNA-FISH. Dashed lines outline representative Leydig cells. Scale bars, 4  $\mu$ m. (B) Luciferase activity in TSL-WT, TSL-*gli1*<sup>-/-</sup>, TSL-*gli2*<sup>-/-</sup> and TSL-*gli3*<sup>-/-</sup> cells transfected with the 8xGLI reporter with or without OnDhh ( $n=4$ ). (C) Volcano plots of transcriptomic changes in TSL-OnDhh, TSL-OnGli1 and TSL+ SAG, compared to TSL-WT. Red and blue dots represent significantly up-and down-regulated genes, respectively. (D) Schematic of the tilapia *sf1* gene promoter, indicating the two predicted Gli1 binding sites (B1 and B2). (E) Transcriptional activation of the *sf1* promoter by Gli1. Competition assays were performed using unlabeled cold probe (Cold probe, GACCACCA, 10/100/250 ng/mL) or mutant unlabeled cold probe (mutant Cold probe, TTAATTA, 10/100/250 ng/mL) ( $n=4$ ). "+" indicates the addition of the corresponding substance, while "-" indicates no addition, and the number represents the amount added (ng/ml). (F-G) Representative immunofluorescence images of testicular sections from recipient fish transplanted with *sf1*-deficient (TSL-*sf1*<sup>-/-</sup>, F) or Sf1-overexpressing (TSL-OnSf1, G) SLCs, stained for Cyp11c1 (green) and PKH26 (red). Nuclei are stained with DAPI (blue). Scale bars, 4  $\mu$ m. (H) Quantification of the percentage of Cyp11c1-positive cells among the transplanted PKH26-positive SLCs in the two transplantation groups ( $n=5$  fish per group). Statistical significance was determined by one-way ANOVA followed by Tukey's test (E, different letters above the error bar indicate statistical differences at  $P < 0.05$ ) or Student's *t*-test (B, H) (\*,  $P < 0.05$ ; \*\*,  $P < 0.01$ ; NS, no significant difference).

oligonucleotide containing the Gli-binding motif (GACCACCCA) competitively inhibited Gli1-driven *sf1* transactivation in a dose-dependent manner, whereas a mutated version (TTAATTAAA) had no effect (Fig. 5E [↗](#)).

The functional necessity of Sf1 was confirmed by transplantation assays using *sf1*<sup>-/-</sup> and Sf1-overexpressing (TSL-*OnSf1*) TSL cells. While TSL-*sf1*<sup>-/-</sup> cells failed to differentiate into Cyp11c1-positive cells even in WT testes, TSL-*OnSf1* cells successfully differentiated and expressed Cyp11c1 even in the *dhh*<sup>-/-</sup> testicular environment (Fig. 5F–H [↗](#)). These results definitively position Sf1 as the critical downstream effector executing the Dhh-Ptch2-Gli1 signaling cascade during SLCs differentiation.

## Discussion

This study delineates a critical Dhh-Ptch2-Gli1-Sf1 signaling axis essential for SLCs differentiation in Nile tilapia. By combining *dhh*<sup>-/-</sup> XY models with SLCs transplantation, we definitively demonstrate that Dhh signaling is dispensable for Leydig cell lineage survival but indispensable for its differentiation. Furthermore, our integrated evidence from targeted gene knockout and functional assays suggests that Ptch2 acts as the primary receptor for Dhh signaling in SLCs, and points to Gli1 as the key transcriptional effector mediating the transactivation of *sf1*. Collectively, these findings implicate Sf1 as the critical downstream target bridging Dhh signaling to steroidogenic maturation, thereby resolving key ambiguities in Leydig cell lineage development and providing evolutionary insights into vertebrate reproductive biology.

Consistent with its established role in mammals, our data in Nile tilapia confirm that Dhh function in male testis development is evolutionarily conserved across vertebrates. While mutations in *Dhh* lead to testosterone insufficiency and infertility in mice [\[22\]](#) and are associated with 46,XY DSD in humans [\[23\]](#), the precise cellular mechanism remained unresolved. Our study demonstrates that *dhh*<sup>-/-</sup> tilapia phenocopy these defects, exhibiting testicular dysgenesis, a profound deficiency of Leydig cells, and drastically reduced 11-KT levels (Fig. 1 [↗](#)). To define the onset of Leydig cell differentiation, we performed a developmental time-course analysis. This revealed that Cyp11c1-positive steroidogenic cells first appear in wild-type testes at 30 dah, while being conspicuously absent in *dhh*<sup>-/-</sup> mutants at this same stage (Fig. S7 [↗](#)). This clear temporal pattern establishes ~30 dah as the developmental window when SLCs initiate their differentiation program in the Nile tilapia. The functional indispensability of Dhh at this specific stage was further confirmed by rescue experiments, in which both SAG treatment and transplantation of Dhh-activated TSL cells restored Cyp11c1-positive cell populations in *dhh*<sup>-/-</sup> testes (Fig. 3 [↗](#)). Collectively, our findings position Dhh as a crucial niche signal that acts at a defined developmental checkpoint to drive the differentiation of SLCs into steroidogenic Leydig cells.

The functional divergence of Ptch1 and Ptch2 in Hh signaling represents an intriguing aspect of pathway regulation. Structurally, both receptors share conserved 12-transmembrane domains vital for cholesterol transport but diverge in their cytoplasmic C-terminal regions, which are implicated in differential Smoothened inhibition efficiency [\[37, 38\]](#). In mammals, Ptch1 serves as the primary, broad-spectrum regulator of Hh signaling during embryogenesis, whereas Ptch2 often assumes context-dependent roles, such as in primordial germ cell migration via its co-receptor Gas1 [\[27\]](#). However, their potential redundancy or specificity within the Leydig cell lineage has remained an open question. Our study provides significant insight into this issue. We confirmed that both *ptch1* and *ptch2* are expressed in Leydig cells (Fig. 4A [↗](#)). Functional interrogation using TSL cell lines revealed a marked difference: genetic ablation of *ptch2* profoundly attenuated Dhh-induced signaling, as measured by a Gli-responsive luciferase reporter, whereas loss of *ptch1* had no detectable effect under the same conditions (Fig. 4B–D [↗](#)). This *in vitro* data, pointing to a primary role for Ptch2, was further supported by *in vivo* genetic interaction studies. The observation that the *dhh*<sup>-/-</sup> phenotype was fully rescued in *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> double mutants (Fig. 4E–U [↗](#)) is consistent with Ptch2 acting as the inhibitory receptor whose loss alleviates pathway suppression. The apparent specificity for Ptch2 in this context might stem from unique co-receptor interactions or expression patterns within the testicular niche. Nonetheless, the early embryonic lethality of global *ptch1* mutation in tilapia [\[39\]](#) precluded a direct assessment

of its potential role in postnatal testis development. Therefore, while our findings strongly nominate Ptch2 as the principal receptor for Dhh in SLCs, a definitive exclusion of a role for Ptch1 will require future studies employing Leydig cell-specific conditional knockout models.

Gli is a nuclear transcription factor, characterized by the presence of a zinc finger domain located in the middle of the protein [40]. Downstream of the zinc finger domain is a phosphorylation cluster containing phosphorylation sites for protein kinase A and glycogen synthase kinase 3, which negatively regulate the transcriptional activity of Gli through phosphorylation [41]. In *Drosophila*, Ci is the sole Gli homolog, possessing both transcriptional activation and repression functions [42]. In vertebrates, there are three Gli homologs: Gli1, Gli2, and Gli3. Gli1 functions solely as a transcriptional activator because it lacks proteolytic processing domains. Gli2 has both activation and repression functions, while Gli3 primarily acts as a transcriptional repressor [43, 44]. However, there has been no research on the regulatory mechanism of Gli1/2/3 in SLCs differentiation. In this study, we found that although *gli1*, *gli2*, and *gli3* are all expressed in Leydig cell lineage (Figure 5A-C), the results from TSL cell lines with mutations in *gli1*, *gli2*, and *gli3* indicated that TSL-*gli1*<sup>-/-</sup>, rather than TSL-*gli2*<sup>-/-</sup> or TSL-*gli3*<sup>-/-</sup>, failed to respond to Dhh signaling (Figure 5D). This suggests that Gli1 is the key transcription factor downstream of the Dhh signaling cascade in SLCs.

Sf1 and Dhh play important roles in Leydig cell commitment and testicular development [5, 11]. Studies suggest a potential interaction between these two factors. For example, in *Dhh*<sup>-/-</sup> XY mice, testicular development is impaired, with a significant reduction in the number of Leydig cells and a marked downregulation of *Sf1* expression [26]. In XX mouse ovaries where Smo is constitutively activated, a large number of ectopic Sf1-positive Leydig cells are generated [45]. Similarly, in *sf1*<sup>-/-</sup> XY tilapia, testicular development is impaired, with a significant reduction in Leydig cell numbers and down regulation of *dhh* expression [10]. These findings imply that Dhh and Sf1 may interact through feedback loops or co-regulatory mechanisms. However, direct evidence for the regulation of Sf1 by the Hh signaling pathway remains lacking. This study elucidates the molecular mechanism of Sf1 as a critical downstream effect of the Dhh-Ptch2-Gli1 axis. Transcriptomic profiling revealed an increased expression of *sf1* by Dhh regulation signaling in TSL cells (Fig. S6), identifying it as a key downstream target of Dhh signaling. Promoter analysis identified two conserved Gli-binding motifs (Fig. 5F) within the *sf1* promoter, and luciferase assays confirmed that Gli1 transactivates *sf1* transcription (Fig. 5G). Functional validation demonstrated that TSL-*sf1*<sup>-/-</sup> cells failed to differentiate even in WT testes, whereas TSL-OnSf1 cells restored Leydig cell differentiation and steroidogenesis in *dhh*<sup>-/-</sup> testes (Fig. 5H-J). These results establish Sf1 as both necessary and sufficient for Leydig cell lineage differentiation.

In conclusion, our study systematically deciphers the signaling cascade governing SLCs differentiation, from the niche signal to the core steroidogenic regulator. We definitively show that the Dhh signal from the niche is indispensable for SLCs differentiation, but not their survival. Within this pathway, Ptch2 serves as the critical receptor, relaying the signal through the transcription factor Gli1. The pivotal outcome of this cascade is the activation of *sf1* by Gli1. Thus, we establish a complete Dhh-Ptch2-Gli1-Sf1 axis, resolving the long-standing question of how a key morphogen controls Leydig cell lineage development in vertebrates.

## Materials and methods

### Animals

Nile tilapias (*Oreochromis niloticus*) were kept in recirculating freshwater systems at 26°C under a 12-h light/dark photoperiod. All animal experiments were conducted in accordance with the regulations of the Guide for Care and Use of Laboratory Animals prescribed by the Committee of Laboratory Animal Experimentation at Southwest University, China (IACUC-20181015-12).

## Generation of mutant lines by CRISPR/Cas9 in Nile tilapia

The sequences of tilapia *dhh* (GenBank ID: 100707022) and *ptch2* (GenBank ID: 100692939) were retrieved from the NCBI. CRISPR/Cas9-mediated mutagenesis was performed as described previously [46]. Guide RNAs (gRNAs) targeting the first exon of *dhh* (GCGGGCCCGTCCGCATCCC) and *ptch2* (GTCCCAGGGGCCGCGTATT) were designed using ZiFit (<http://zifit.partners.org/ZiFit/>). gRNA and Cas9 mRNA were synthesized following established protocols [46]. The primers listed in Table S1 [46]. A mixture of synthetic gRNA (500 ng/μl) and Cas9 mRNA (1000 ng/μl) was microinjected into one-cell-stage fertilized embryos. Mutants were validated by polyacrylamide gel electrophoresis (PAGE) and Sanger sequencing. F0 chimeric XY males were crossed with WT XX females to generate heterozygous F1 offspring. F1 siblings carrying a 13-bp deletion in *dhh* or a 25-bp deletion in *ptch2* were intercrossed to generate homozygous F2 mutants. For double mutants, *dhh*<sup>+/-</sup> XY males were mated with *ptch2*<sup>+/-</sup> XX females to obtain *dhh*<sup>+/-</sup>;*ptch2*<sup>+/-</sup> offspring. These double heterozygotes were then intercrossed to generate *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> mutants.

## Generation of mutant lines by pCas9-NtU6sgRNA system in TSL Cells

The tilapia stem Leydig cell line TSL was maintained in ESM4 medium as described [33]. Gene sequences for *ptch1* (GenBank ID: 100700194), *ptch2* (GenBank ID: 100692939), *gli1* (GenBank ID: 100710687), *gli2* (GenBank ID: 100704788), *gli3* (GenBank ID: 100707331), and *sf1* (GenBank ID: 100534561) were retrieved from NCBI. Knockout plasmids targeting these genes were designed following established protocols [47]. gRNAs were generated using ZiFit, with the following exon-specific targets: *ptch1* (GGAGGCGCTCCTGCAGCACC, exon 3), *ptch2* (GTCCCAGGGGCCGCGTATT, exon 1), *gli1* (TGACTCATGGATCAGGACCA, exon 2), *gli2* (GATTCGCTGTACCCACG, exon 3), *gli3* (TGAGGAGCCCTCTACGTCTA, exon 2), and *sf1* (GGCTGTGTACTGGTACTGGG, exon 4; designed as previously described) [47]. Primers are listed in Table S1 [46]. Plasmids were transfected into TSL cells using the MDMP E-1 microchannel transfection instrument (MENDGENE) with transwell cell compartments (BIOFIT), per manufacturer guidelines. Post-transfection (48 hours), cells were selected in ESM4 medium containing 500 μg/mL G418 for 7 days. Single-cell clones were isolated via dilution and serial passage, followed by genomic DNA (gDNA) extraction and Sanger sequencing to confirm mutations.

## Histological analyses of testes from WT, *dhh*<sup>-/-</sup>, *ptch2*<sup>-/-</sup> and *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> XY fish

GSI was calculated for WT, *dhh*<sup>-/-</sup>, *ptch2*<sup>-/-</sup> and *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> XY fish at 90 dah as: (gonad weight / body weight) × 100%. Testes were fixed in Bouin's solution for 24 hours at room temperature with agitation, dehydrated through an ethanol series, and embedded in paraffin. Sections (5 μm thickness) were stained with hematoxylin and eosin (H&E) and imaged using an Olympus BX53 microscope. Germ cell classification (spermatogonia, spermatocytes) followed established criteria [48].

## Immunofluorescence

Testes from WT, *dhh*<sup>-/-</sup>, WT+11-KT, *dhh*<sup>-/-</sup>+11-KT, *ptch2*<sup>-/-</sup> and *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> XY fish at 90 dah were analyzed for germ cell marker Vasa, Leydig cell marker Cyp11c1, and meiosis marker Sycp3. Tissue fixation and embedding. Sections (5 μm) were deparaffinized, rehydrated, and subjected to antigen retrieval. After blocking with 5% bovine serum albumin (BSA; 30 minutes, room temperature), sections were incubated overnight at 4°C with rabbit polyclonal primary antibodies (Vasa, Cyp11c1, Sycp3; 1:1,000 dilution), validated previously [49]. Alexa Fluor 488-conjugated goat anti-rabbit secondary antibody (1:1,000; Invitrogen) was applied for 1 hour at 37°C, followed by DAPI nuclear counterstaining (1:1,000; Sigma-Aldrich). Fluorescence signals were imaged using an Olympus Fv3000 confocal microscope. For each biological replicate (*n*=5-6 fish per genotype), three



non-serial, non-adjacent testis sections were analyzed. From each section, three representative fields of view were captured to ensure non-overlapping sampling. All positive cells number of Vasa, Sycp3 and Cyp11c1 was quantified by Image J Pro 1.51 software using default parameters.

## Fluorescence *in situ* hybridization (FISH) and immunofluorescence

Testes from WT XY tilapia at 90 dah were fixed in 4% paraformaldehyde (Sigma) overnight at 4°C with agitation, paraffin-embedded, and sectioned (5 µm). FISH was performed as described [48]. Partial cDNA fragments of *ptch1*, *ptch2*, *gli1*, *gli2*, and *gli3* were cloned into the pGEM-T Easy vector. Sense (control) and antisense RNA probes were synthesized using a DIG RNA Labeling Kit (Roche) and T7 mMESSAGE mMACHINE Kit (Ambion). Signals were amplified with the TSA<sup>TM</sup> Plus TMR system (PerkinElmer) per manufacturer protocols (primers in Table S1 [48]). Concurrently, Leydig cell marker Cyp11c1 expression was assessed by immunofluorescence.

## 11-KT and SAG rescue experiments

To rescue *dhh*<sup>-/-</sup> XY fish, offspring from *dhh*<sup>+/-</sup> XY male × *dhh*<sup>+/-</sup> XX female crosses were divided into two groups. The experimental group received 11-KT via immersion from 30 to 90 dah, with treatments refreshed every 2 days. 11-KT concentrations increased incrementally from 57 ng/L at 30 dah to 1254 ng/L by 90 dah (39.9 ng/L increase per treatment cycle; see Fig. 2 [48], 5 [48], S7 [48]). For SAG rescue, fish received intraperitoneal injections of 10 mg/kg SAG from 30 to 90 dah, with booster doses administered weekly. Testes were collected at 90 dah for morphological and histological evaluation.

## TSL cell labeling, transplantation and IF

TSL, previously isolated from 3-month-old XY tilapia testes [33], was used to assess stem Leydig cell differentiation in WT and *dhh*<sup>-/-</sup> XY fish. TSL cells were treated as follows: (1) untreated TSL, (2) TSL + 0.5 µM SAG (Hh pathway agonist), (3) TSL-*OnDhh* (overexpressing tilapia Dhh), (4) TSL-*OnSf1* (overexpressing tilapia Sf1), and (5) TSL-*sf1*<sup>-/-</sup> (*sf1* mutation). For the SAG treatment experiment, TSL cells were incubated with 0.5 µM SAG for 48 hours before transplantation. Cells were labeled with PKH26 fluorescent dye (Sigma) and transplanted (5,000 cells per testis) into WT or *dhh*<sup>-/-</sup> XY tilapia via the urogenital tract [33]. For TSL-*OnDhh*, the Dhh ORF (1377 bp) was amplified from gDNA, cloned into pcDNA3.1 (Invitrogen), and transfected into TSL cells (primers in Table S1 [48]). At 10 days post-transplantation, testes were analyzed by IF to quantify Cyp11c1 positive cells among PKH26 positive populations.

## GLI report assay

A firefly luciferase reporter plasmid (8×GLI) was generated by cloning eight tandem copies of the GLI-binding motif (GACCACCA) into the pGL4.23 vector (minimal promoter). To validate 8×GLI functionality, TSL cells were transiently transfected with: (1) pGL4.23 (negative control), (2) 8×GLI, or (3) 8×GLI plus Dhh-overexpression plasmid (*OnDhh*), for 48 hours. The 8×GLI construct was further transfected into TSL (WT), TSL-*ptch1*<sup>-/-</sup>, TSL-*ptch2*<sup>-/-</sup>, TSL-*gli1*<sup>-/-</sup>, TSL-*gli2*<sup>-/-</sup> or TSL-*gli3*<sup>-/-</sup>, with or without *OnDhh*. Transfections included pRL-TK as an internal control. Luciferase activity was measured 48 hours post-transfection using the Dual Luciferase Assay System (Promega) and a Luminoskan<sup>TM</sup> Ascent luminometer (Thermo Fisher Scientific). Relative activity was calculated as firefly/Renilla luciferase ratios.

## Western blot

The ORFs of Dhh (1377 bp) and Gli1 (4404 bp) were amplified from gDNA by PCR. Two tandem FLAG tag sequences (GATTATAAAGATGATGATAAA) were inserted flanking each ORF (after the start codon and before the stop codon) and cloned into the pcDNA3.1 vector (Invitrogen), generating pcDNA3.1-*OnDhh* and pcDNA3.1-*OnGli1* (primers in Table S1 [48]). TSL cells were transiently transfected with pcDNA3.1-*OnDhh* or pcDNA3.1-*OnGli1* for 48 hours. Protein expression was analyzed by Western blot following established methods [49]. Briefly, total protein lysates were separated by 15% SDS-PAGE, transferred to nitrocellulose membranes, and blocked

with 5% BSA in TBST (10 mM Tris pH 7.9, 150 mM NaCl, 0.1% Tween-20) for 1 hour at 37°C. Membranes were incubated overnight at 4°C with rabbit anti-FLAG antibody (1:1000; Cell Signaling Technology), followed by HRP-conjugated goat anti-rabbit secondary antibody (1:1000; Invitrogen) for 1 hour at 37°C. Signals were detected using the BeyoECL Plus Kit (Beyotime) and imaged on a Fusion FX7 system (Vilber Lourmat).

## Measurement of 11-KT levels by ELISA

Blood was collected from the caudal vein of 90 dah XY fish (WT, *dhh*<sup>-/-</sup>, *ptch2*<sup>-/-</sup>, *dhh;ptch2*<sup>-/-</sup>, WT+TSL, *dhh*<sup>-/-</sup>+TSL, *dhh*<sup>-/-</sup>+TSL+SAG, *dhh*<sup>-/-</sup>+TSL-*OnDhh* and *dhh*<sup>-/-</sup>+TSL-*sf1*<sup>-/-</sup>), incubated at 4°C overnight, and centrifuged to isolate serum. Tissue fluid samples from WT and *dhh*<sup>-/-</sup> XY fish at 5, 10, 20, and 30 dah were similarly prepared. All samples were stored at -80°C until analysis. Serum and tissue fluid 11-KT levels were quantified using a 11-keto Testosterone ELISA Kit (Cayman) validated for teleost samples, per manufacturer instructions. Individual samples (serum or tissue fluid; *n* = 6 per genotype) were analyzed in triplicate on a single 96-well plate, with concentrations calculated from standard curves.

## Transcriptome analyses

Tissue- and stage-specific expression of *dhh* was assessed using transcriptomic data from tilapia tissues (brain, heart, head kidney, kidney, liver, muscle, ovary, testis) and gonads at developmental stages spanning 5 to 360 dah [50, 51]. For global gene expression profiling, RNA was extracted from TSL-WT, TSL-*OnDhh*, TSL-*OnGli1*, and TSL + 0.5 μM SAG cells using RNAiso Plus (Takara). For the SAG treatment experiment, TSL cells were incubated with 0.5 μM SAG for 48 hours before collection. For each genotype, cells from three independent culture wells were pooled. Libraries were sequenced on an Illumina platform (GENEBOOK Biotechnology), and clean reads were aligned to the reference genome with HISAT2. Gene expression levels (FPKM: fragments per kilobase per million mapped reads) were quantified using featureCounts. Differentially expressed genes (DEGs) were identified for each condition (TSL-*OnDhh*, TSL-*OnGli1*, TSL+SAG) compared to TSL-WT controls using edgeR (threshold: FDR < 0.05, |log2(foldchange)| ≥ 1.5).

## Real-time PCR

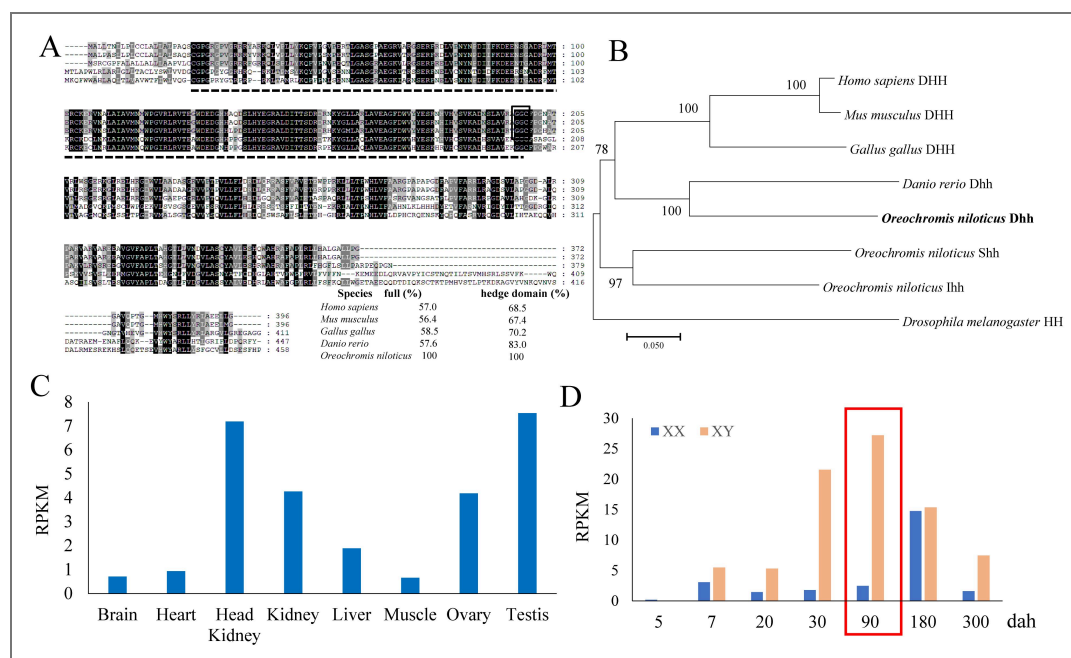
Testes from WT and *dhh*<sup>-/-</sup> XY fish at 90 dah (*n* = 3 per genotype), along with TSL-WT, TSL-*OnDhh*, TSL-*OnGli1*, and TSL + 0.5 μM SAG cells, were analyzed for gene expression. Total RNA was extracted using RNAiso Plus (Takara), reverse-transcribed into cDNA (PrimeScript II 1st Strand cDNA Synthesis Kit; Takara), and amplified by real-time PCR on an ABI-7500 system (Applied Biosystems). Relative mRNA levels were normalized to β-actin and calculated via the formula  $R = 2^{-\Delta\Delta C_t}$  (primers in Table S1 [4]).

## Dual luciferase report assay

The 5015-bp promoter region upstream of the *sf1* start codon was amplified and cloned into the pGL3-basic vector (Promega), generating pGL3-*sf1* (primers in Table S1 [4]). To assess Gli1 binding to the *sf1* promoter, HEK293 cells were co-transfected with pGL3-*sf1*, pcDNA3.1-*OnGli1*, cold probe (GACCACCCA), and mutant cold probe (TTAATTAAA) using the TransIT-X2 transfection reagent (Mirus Bio). Control transfections included empty pcDNA3.1 and pGL3 vectors, with pRL-TK for normalization. Luciferase activity was measured 48 hours post-transfection.

## Statistical analyses

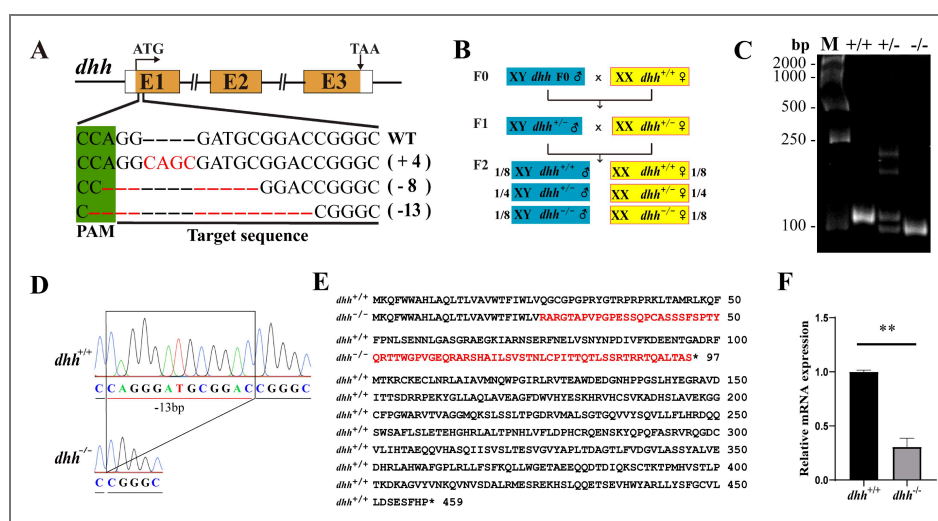
Data are presented as mean ± SD from at least three independent replicates. A two-tailed independent Student's t-test was used to determine the differences between the two groups. One-way ANOVA, followed by Tukey multiple comparison, was used to determine the significance of differences in more than two groups. *P* < 0.05 was used as a threshold for statistically significant differences.



**Fig. S1. Sequence analysis and mRNA expression profile of Nile tilapia *dhh*.** (A) Dhh amino acid sequence alignment. Amino acids are numbered in the right margin. Deletions are indicated by dashes, shaded areas indicate shared sequences. The amino-terminal hedge domain is underlined by dotted line, with the signal peptide region in the front and the autocatalytic carboxy-terminal domain (hog) in the back. The box indicates the autocatalytic site of an absolutely conserved Gly-Cys-Phe tripeptide. At the end of the alignment are percentage identity values of the full-length and hedge domain of tilapia Dhh to the orthologue from other species. (B) Phylogenetic analysis of Dhh. The phylogenetic tree was constructed using the neighbor-joining method within the MEGA7.0 program. Node values represent percent bootstrap confidence derived from 1000 replicates. (C-D) The RPKM (reads per kb per million reads) values of tilapia *dhh* in various adult tissues (C) and XY and XX gonads at 5, 7, 20, 30, 90, 180, 300 dah (days after hatching) (D) in transcriptome sequencing data which were sequenced using Illumina 2000 HiSeq technology in our previous study (Tao et al., 2013).

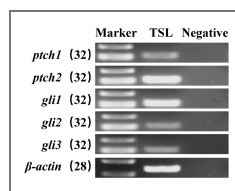
**Fig. S2. Establishment of the Nile tilapia *dhh* mutant line by CRISPR/Cas9.**

(A) Schematic representation of gRNA targeting the Nile tilapia *dhh* locus. The gRNA was designed to target exon 1. The translation start codon ATG and stop codon TAA were indicated by arrows. The PAM (protospacer adjacent motif) site was marked by green box. DNA sequence alignment of the three mutant lines with WT. The added sequences were marked by red letter, and the deleted sequences were indicated in red dotted lines. One had a 4-bp addition, one had an 8-bp deletion, and the other had a 13-bp deletion. (B) Schematic diagram showing the breeding plans of *dhh* F0 to F2 fish. (C) Homozygous mutants of F2 fish were screened by PAGE. The first lane is DNA marker, the second lane is *dhh*<sup>+/+</sup> fish, the third lane is *dhh*<sup>+/-</sup> fish, and the fourth lane is *dhh*<sup>-/-</sup> fish. (D) Sanger sequencing results of *dhh* genes from WT and the 13-bp deletion homozygous mutant fishes. (E) The *dhh* ORF sequence in WT was indicated in black and the frameshift-altered sequence for the 13-bp deletion *dhh* ORF was indicated in red. (F) The relative mRNA expression of *dhh* in WT and the 13-bp deletion homozygous mutant fishes by RT-qPCR. Values were presented as mean  $\pm$  SD ( $n=3$ ). Differences were determined by two-tailed independent Student's t-test. \*\*,  $P < 0.01$ .



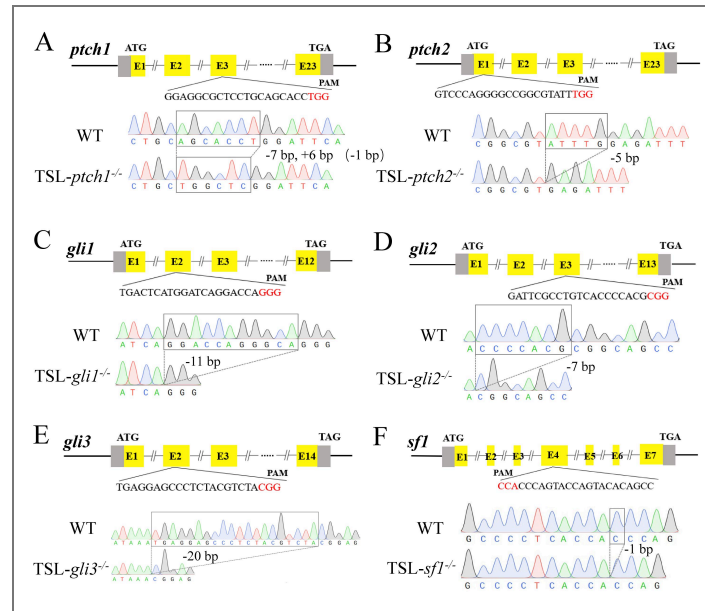
**Fig. S3. RT-PCR analyses of Hh pathway genes in TSL cells.**

$\beta$ -actin served as a loading control, and RNAs from TSL cells were used as templates for negative controls. The numbers in parentheses indicate the PCR cycles.



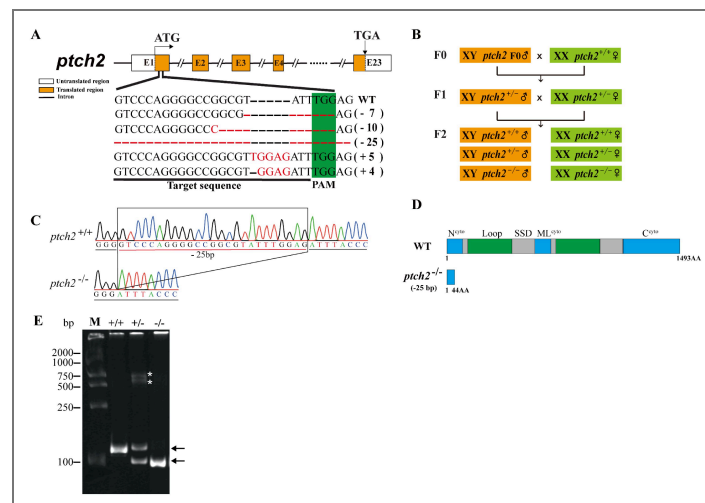
**Fig. S4. Establishment of the TSL *ptch1*, *ptch2*, *gli1*, *gli2*, *gli3* and *sf1* mutant lines.**

(A-F) Schematic representation of gRNAs targeting the *ptch1*, *ptch2*, *gli1*, *gli2*, *gli3* and *sf1* locus. The gRNAs were designed to target ORF (open reading frame). The PAM (protospacer adjacent motif) sites were marked by red letter. Sanger sequencing of *ptch1*, *ptch2*, *gli1*, *gli2*, *gli3* and *sf1* single mutant alleles in single cell clones. PCR amplicons from DNA templates of the six cell clones were directly used for sequencing.



**Fig. S5. Establishment of the Nile tilapia *ptch2* mutant line by CRISPR/Cas9 system.**

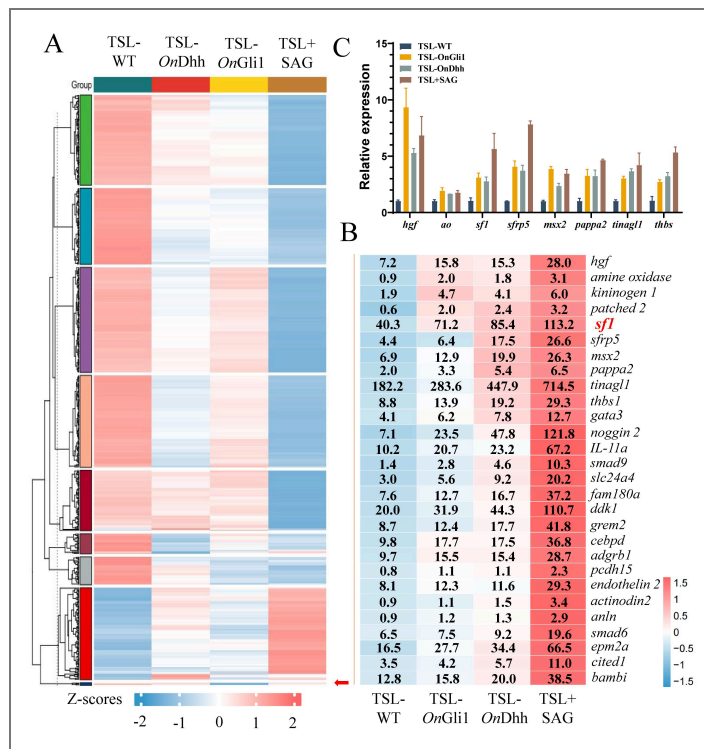
(A) Schematic representation of gRNA targeting the Nile tilapia *ptch2* locus. The gRNA was designed to target exon 1. The translation start codon ATG and stop codon TGA were indicated by arrows. The PAM site was marked by green box. DNA sequence alignment of the five mutant lines with WT. The added sequences were marked by red letter, and the deleted sequences were indicated in red dotted lines. (B) Schematic diagram showing the breeding plans of *ptch2* F0 to F2 fish. (C) DNA sequencing showed the 25-bp deletion within the *ptch2* ORF in the homozygous mutant compared to the WT. (D) Schematic of the prediction of the intact Ptc2 protein in the WT and the truncated Ptc2 protein in the homozygous mutants. (E) Homozygous mutants of F2 fish were screened by PAGE. The first lane is DNA marker, the second lane is *ptch2*<sup>+/+</sup> fish, the third lane is *ptch2*<sup>-/-</sup> fish, and the fourth lane is *ptch2*<sup>-/-</sup> fish. “\*” indicates heteroduplex, and arrows indicate homoduplex.





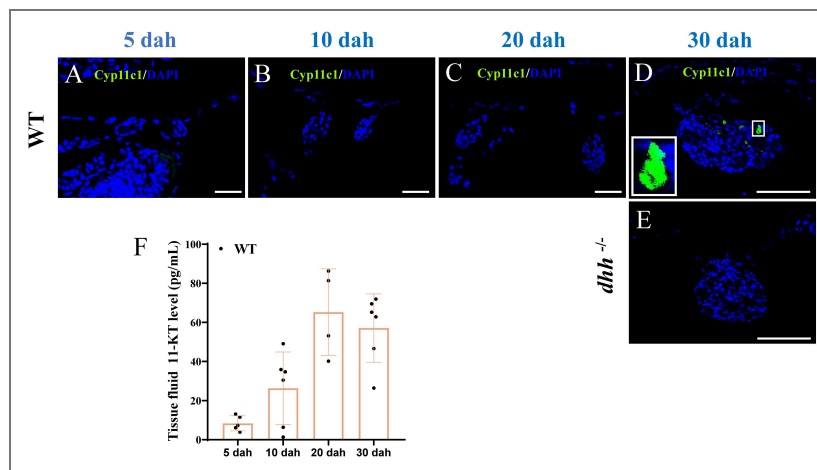
**Fig. S6. Transcriptome data analyses and verification by RT-qPCR.**

(A) Hierarchical clustering analysis of global gene expression patterns in TSL-WT, TSL-*OnDhh*, TSL-*OnGli1* and TSL+ 0.5  $\mu$ M SAG. Blue indicates decreased expression, and red indicates increased expression. (B) Heatmap showing the expression patterns of upregulated differentially expressed genes (DEGs) identified in Fig. 5C. The FPKM value for each gene in each sample is indicated within the squares. The color gradient from blue to red reflects low to high expression levels per row (gene). (C) qRT-PCR validation. Values were presented as mean  $\pm$  SD ( $n = 3$ ).



**Fig. S7. The expression levels of Cyp11c1 in the testis and the levels of 11-KT in tissue fluid from WT and *dhh*<sup>-/-</sup> XY fish at 5, 10, 20, and 30 dah.**

(A-E) Representative immunofluorescence images showing Cyp11c1 expression (green) in testis sections from WT fish at 5, 10, 20, 30 dah and *dhh*<sup>-/-</sup> mutants at 30 dah. Scale bars, 4  $\mu$ m. (A) (F) Tissue fluid 11-KT levels in WT and *dhh*<sup>-/-</sup> XY fish at indicated time points ( $n=6$  fish per group). Values were presented as mean  $\pm$  SD. Differences were determined by two-tailed independent Student's t-test. NS, no significant difference.



## Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request

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## Additional information

### Author contributions

**Changle Zhao:** Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Yongxun Chen:** Writing – original draft, Methodology, Data curation. **Lei Liu:** Data curation, Conceptualization. **Xiang Liu:** Software, Investigation. **Hesheng Xiao:** Software, Data curation. **Feilong Wang:** Investigation, Formal analysis. **Qin Huang:** Resources, Software. **Xiangyan Dai:** Resources, Investigation. **Wenjing Tao:** Resources, Funding acquisition. **Deshou Wang:** Supervision, Investigation, Funding acquisition. **Jing Wei:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Data curation, Conceptualization.

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## Additional files

[Table S1](#) 

[Graphical abstract](#) 

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## Peer reviews

### Reviewer #1 (Public review):

#### Summary:

This manuscript by Zhao et. al investigates the canonical hedgehog pathway in testis development of Nile tilapia. They used complementary approaches with genetically modified tilapia and transfected TSL cells (a clonal stem Leydig cell line) previously derived from 3-month old tilapia. The approach is innovative and provides a means to investigate DHH and each downstream component from the ptch receptors to the gli and sf1 transcription factors. They concluded that Dhh binds Ptch2 to stimulate Gli1 to promote an increase in Sf1 expression leading to the onset of 11-ketotestosterone synthesis heralding the differentiation of Leydig cells in the developing male tilapia.'

#### Strengths of the methods and results:

- The use of Nile tilapia is important as it is an important aquaculture species, it shares the genetic pathway for sex determination of mammalian species, and molecular differentiation pathways are highly conserved
- The approach is rigorous and incorporates a novel TSL, clonal stem Leydig cell model that they developed that is relatively faithful in following endogenous developmental steps and can produce the appropriate steroid.
- Tilapia are relatively amenable to CRISPR/Cas9 targeting and, with their accelerated developmental time frame, provide an excellent model system to interrogate specific signaling pathways.
- The stepwise analysis from dhh-gli-sf1 is thoughtful and well done.



#### Weaknesses of the methods and results:

- Line 162: need to establish and verify the PKH26-labeled TSL cells were unaffected by the *dhh*<sup>-/-</sup> environment. No data to support the claim that they were unaffected.
- The rescued phenotype caused by the addition of *ptch2*<sup>-/-</sup> to the *dhh*<sup>-/-</sup> model is a compelling. To further define potential *ptch1* contributions, it would be helpful to examine the expression level of *ptch1* in the context of the *ptch2*<sup>-/-</sup> and *ptch2*<sup>-/-</sup>;*dhh*<sup>-/-</sup> mutant animals. Any compensatory increase in *ptch1* in either case, without obvious phenotype changes, would support the dominant role for *ptch2*.
- Activity of individual gli factors need additional reconciliation. The expression profiles for both alternative gli factors should be quantified in each knockout cell line to establish redundancy and/or compensation.
- Figure 5E: An important control is missing that includes evaluation of HEK293 cells transfected with pCDNA3.1-OnGli1 without the addition of pGL3-sf1.

#### Achieved Aims:

The authors set out to test the hypothesis that the canonical Dhh signaling pathway for Leydig cell differentiation and steroidogenic activity is mediated via *ptch2* and *gli1* regulation of *sf1*. The results are strong, there are additional steps needed to verify that redundancy/compensation is not contributing to the outcomes.

This work is important in better understanding of nuanced commonalities and differences in developmental pathways across species. Specific to Leydig cell differentiation and steroidogenesis, their work with tilapia supports conservation of the canonical Dhh pathway; however, there appear to be some differences in downstream mediators compared to mouse. Specifically, they conclude that *ptch2*/*gli1* stimulates *sf1* and steroidogenesis in tilapia where *gli1* is dispensable in mouse. Instead, *Gli3* has recently been shown to play an important role to stimulate *Sf1* and support the hedgehog pathway.

<https://doi.org/10.7554/eLife.109979.1.sa1>

## Author response:

### General Statements

We thank the reviewers for their thoughtful and constructive comments on our manuscript. We have thoroughly considered all points raised and have made extensive revisions to address them. These revisions have significantly strengthened the manuscript.

In summary, the key revisions and clarifications include:

(1) Developmental Time-Course: To address the need for earlier phenotypic analysis, we have performed new immunofluorescence experiments at 30 days after hatching (dah). This new data (Fig. S7) precisely pinpoints the onset of the Leydig cell differentiation defect in *dhh*<sup>-/-</sup> mutants, establishing ~30 dah as the critical window for Dhh action.

(2) Role of *Ptch1* and *Ptch2*: We have qualified our conclusions regarding receptor specificity throughout the text to accurately reflect our findings and the limitation posed by the early lethality of *ptch1* mutants. The in vivo genetic evidence for *Ptch2* (the rescue of *dhh*<sup>-/-</sup> by *ptch2*<sup>-/-</sup>) is emphasized, while we now explicitly state that a role for *Ptch1* cannot be ruled out without future conditional knockout models.

(3) Mechanism between *Gli1* and *Sf1*: In direct response to the reviewers' request for stronger evidence, we have performed a new cold probe competition assay. This experiment provides

dose-dependent, biochemical evidence for the specificity of Gli1 binding to the *sf1* promoter (New Fig. 5E). Furthermore, we have revised the text throughout the manuscript to use more precise language (e.g., "Gli1 activates *sf1* expression") and removed overstated claims of "direct" regulation.

(4) Methodological Rigor and Controls: We have added crucial negative controls for all RNA-FISH experiments using sense probes (New Fig. S9), provided detailed quantification methods for immunofluorescence, clarified the number of biological replicates for transcriptomic analyses, and corrected statistical tests as recommended.

(5) Clarity and Presentation: We have revised the text for clarity, expanded the description of the TSL cell line's validation in the Introduction, added missing details to figure legends and methods, and incorporated suggested key references.

We believe that our detailed responses and the significant new data and textual revisions have fully addressed the reviewers' concerns and have substantially improved the quality and impact of our manuscript.

### Point-by-point description of the revisions

#### **Reviewer #1 (Evidence, reproducibility and clarity):**

##### *Summary*

*This manuscript by Zhao et. al investigates the canonical hedgehog pathway in testis development of Nile tilapia. They used complementary approaches with genetically modified tilapia and transfected TSL cells (a clonal stem Leydig cell line) previously derived from 3-mo old tilapia. The approach is innovative and provides a means to investigate DHH and each downstream component from the ptch receptors to the gli and sf1 transcription factors. They concluded that Dhh binds Ptch2 to stimulate Gli1 to promote an increase in Sf1 expression leading to the onset of 11-ketotestosterone synthesis heralding the differentiation of Leydig cells in the developing male tilapia.*

##### *Major comments:*

##### *(1) Are the key conclusions convincing?*

*Most results as reported are convincing; however, some conclusions are premature as additional experiments are required to satisfy their claims. For example, the phenotype of the dhh-/- testis is convincing in that Cyp11c1 cells are missing and the addition of ptch2-/- rescues the phenotype indicating a direct path. The link from gli to sf1, however, requires additional study to validate the direct relationship (see item 3 below).*

We thank the reviewer for the positive assessment that our principal findings are convincing. Regarding the connection between Gli1 and Sf1, we agree that additional validation was important. We have now performed new experiments and revised our text. As detailed in our response to item 3 below, we have incorporated a cold probe competition assay (new Fig. 5E) which provides dose-dependent evidence for the specificity of Gli1 binding to the *sf1* promoter. Furthermore, we have toned down our conclusions in the manuscript.

##### *(2) Should the authors qualify some of their claims as preliminary or speculative, or remove them altogether?*

*Major: Most significant premature claim is the statement that gli1 directly controls sf1 activity. Additional experiments are required to make this claim (see next statement).*

We agree with the reviewer that the claim of "direct" control was premature. We have therefore revised the manuscript accordingly. All statements claiming "direct" regulation of

*sf1* by Gli1 have been removed or replaced with more accurate descriptions, such as "Gli1 activates *sf1* expression" and "Sf1 is a key transcriptional target of Gli1." These changes, coupled with the new functional data from the cold probe competition experiment (Fig. 5E) described in our response to item 3, now provide a robust and appropriately qualified account of our findings.

*Minor: As addressed in the discussion section, the ptch1 animals fail to survive limiting the ability to validate both ptch1 and ptch2 roles. Thus, the conclusion that only ptch2 is required should be qualified.*

We thank the reviewer for this rigorous comment. We fully acknowledge the limitation imposed by the early lethality of *ptch1* mutants, which precludes a definitive *in vivo* assessment of its potential role in postnatal testis development. In direct response to this point, we have revised the text throughout the manuscript to more accurately reflect the strength of our conclusions. Specifically, in the Results section, we now state that "This differential receptor requirement implies that Ptch2 likely acts as the functional receptor for transducing Dhh signals in TSL cells" (lines 174–176). Furthermore, we have strengthened the Discussion by explicitly stating: "Therefore, while our findings strongly nominate Ptch2 as the principal receptor for Dhh in SLCs, a definitive exclusion of a role for Ptch1 will require future studies employing Leydig cell-specific conditional knockout models" (lines 265–268). We believe these revisions provide an appropriately qualified interpretation of our data while maintaining the compelling narrative of Ptch2's primary role.

*Major: There are a couple of key references missing however, please consider including:*

- Kothandapani A, Lewis SR, Noel JL, Zacharski A, Krellwitz K, Baines A, Winske S, Vezina CM, Kaftanovskaya EM, AgoulNIK AI, Merton EM, Cohn MJ, Jorgensen JS. *PLoS Genet.* 2020 Jun 4;16(6):e1008810. doi: 10.1371/journal.pgen.1008810. eCollection 2020 Jun. PMID: 32497091

- Park SY, Tong M, Jameson JL. *Endocrinology.* 2007 Aug;148(8):3704-10. doi: 10.1210/en.2006-1731. Epub 2007 May 10. PMID: 17495005

We have included the key references: Kothandapani A, et al. (2020). *PLoS Genet.* and Park SY, et al. (2007). *Endocrinology.*

*(3) Would additional experiments be essential to support the claims of the paper? Request additional experiments only where necessary for the paper as it is, and do not ask authors to open new lines of experimentation. Additional experiments are suggested to strengthen the direct connection between gli1 and sf1:*

*Major: Figure 5F shows evidence for increased sf1-luc activity upon co-transfection of OnGli1 in TSL cells. These data would be strengthened with evaluation of the same sf1 promoter that has each/both putative GLI binding sites mutated.*

We thank the reviewer for this insightful suggestion. To further strengthen the evidence for the functional connection between Gli1 and the *sf1* promoter, we have performed a new cold probe competition experiment. Given the potential presence of other unpredicted Gli-binding motifs within the 5-kb *sf1* promoter region and the practical constraints, we employed an alternative, robust biochemical approach. This assay used a wild-type oligonucleotide containing the canonical Gli-binding motif (GACCACCA) as a specific competitor. As shown in the new Fig. 5E, this cold probe caused a significant, dose-dependent reduction in Gli1-induced *sf1*-luc activity, while a mutated control probe (TTAATTAAA) had no effect. This result provides strong evidence that Gli1-mediated transactivation of the *sf1* promoter is dependent on its specific binding to this consensus motif.

Furthermore, in response to the reviewer's comment, we have revised the manuscript text to use more precise language, such as "Gli1 activates *sf1* expression" and "Sf1 is a key transcriptional target of Gli1," toning down any overstated claims of direct regulation. Together with the existing data-which includes the original luciferase assay, the new competition experiment, and key loss-of-function/gain-of-function genetic evidence from SLCs transplantation-we believe our study now provides a compelling and multi-faceted case for Gli1 being the key regulator of *sf1* within this pathway. We are confident that these revisions have satisfactorily addressed the point raised.

*Major: All 8xGli-luciferase assays should include evaluation of the mutant 8xGli-luciferase plasmid as a negative control.*

We thank the reviewer for highlighting the importance of reporter assay controls. In our study, we included the empty vector pGL4.23, which lacks any Gli-binding sites, as the fundamental negative control. As shown in Fig. 4C, this vector showed minimal background activity that was unresponsive to Dhh, confirming that the strong luciferase induction in the 8xGli-reporter is entirely dependent on functional Gli-binding sites. While a mutated 8xGli construct is one valid approach, we think that the use of an empty vector is functionally equivalent and equally rigorous for establishing specificity. We are confident that our current data unambiguously demonstrate Gli-dependent activation. For clarity, we have explicitly stated in the figure legend and methods that pGL4.23 served as the negative control.

*Minor: Figure 5D experiment that includes TSL-gli1(also 2,3) +/- OnDhh; please examine whether the absence of Gli affects expression of sf1 in each condition. In other words, provide a loss-of-function of Gli connection to regulation of sf1.*

We measured the mRNA expression levels of *sf1* in TSL-WT, TSL-*gli1*<sup>-/-</sup>, TSL-*gli2*<sup>-/-</sup>, and TSL-*gli3*<sup>-/-</sup> cells using qRT-PCR. The results are presented in the new Supplementary Figure S8A. The results show that the loss of *gli1* leads to a significant reduction in the expression of *sf1*. In contrast, the knockout of *gli2* or *gli3* had no significant effect on *sf1* expression levels.

*(4) Are the suggested experiments realistic in terms of time and resources? It would help if you could add an estimated cost and time investment for substantial experiments.*

*Given the expertise, it is not anticipated that the suggested experiments would be a significant burden to this group.*

We appreciate the reviewer's considerations. Now, we have performed the additional key experiments, which have been incorporated into the revised manuscript. We believe these new data have fully addressed the points raised.

*(5) Are the data and the methods presented in such a way that they can be reproduced?*

*Most methods are adequately described or referenced to previous detailed description. There were, however, some methods that could benefit from additional details:*

*Major: IF quantification data: please provide details on how the number of positive cells were quantified and presented, for example, how many cells from how many sections for each genotype were included for the analysis?*

We have added relevant information in the "Materials and Methods" section in line 369-373: "For each biological replicate ( $n=5-6$  fish per genotype), three non-serial, non-adjacent testis sections were analyzed. From each section, three representative fields of view were captured to ensure non-overlapping sampling. All positive cells number of Vasa, Sycp3 and Cyp11c1 was quantified by Image J Pro 1.51 software using default parameters."

*Major: FISH: No controls are present, for example, scrambled RNA probes. Further, please clarify or address the significant presence of message in the nucleus.*

As suggested, we have now included negative control experiments using sense RNA probes for all genes (*ptch1*, *ptch2*, *gli1*, *gli2*, *gli3*). These controls showed no specific signal, confirming the specificity of our antisense probe hybridization. These data are now presented in the new Supplementary Figure S9.

*Major: TSL cells: TSL-onDhh, -onSf1: provide evidence for increase in expression*

We measured the mRNA expression levels of *dhh* in TSL-WT and TSL-*OnDhh*, and *sf1* in TSL-WT and TSL-*OnSf1* using qRT-PCR. The results are presented in the new Supplementary Figure S8B. The results show that overexpression of Dhh and Sf1 significantly increased the mRNA expression levels of *dhh* and *sf1*, respectively.

*Major: TSL + SAG cells and other treatments in general: how long were they treated before transplantation?*

Response: We have added relevant information in the "Materials and Methods" section in line 398-399: "For the SAG treatment experiment, TSL cells were incubated with 0.5  $\mu$ M SAG for 48 hours before transplantation."

*Major: Transcriptome analyses: how many replicates were used for each cell line? Please clarify-the results presented in Fig 5E: how was this plot generated, it is interpreted that all three cell lines were combined and compared to the WT line. It is not clear how this was achieved.*

We have added relevant information in the "Materials and Methods" section in line 445-447: "For the SAG treatment experiment, TSL cells were incubated with 0.5  $\mu$ M SAG for 48 hours before collection. For each genotype, cells from three independent culture wells were pooled.

Added relevant information in the "Results" section in line 198-202: "...we performed transcriptomic profiling of TSL cells under conditions of pathway activation: Dhh overexpression (TSL-*OnDhh*), Gli1 overexpression (TSL-*OnGli1*), and SAG treatment (TSL+SAG). Comparative RNA-seq analysis identified a core set of 33 genes consistently upregulated across all three conditions."

*(6) Are the experiments adequately replicated and statistical analysis adequate?*

*Most are adequate and appropriate, some questions remain:*

*- Transcriptomes-how many replicates (see above)?*

*- IF quantification-how were cells identified/how many sections (see above)?*

*Minor: Statistics: methods indicate that a student's t-test was used, but ANOVA's are also used, which is appropriate. There are data presented that should be reevaluated via an ANOVA: Figure 4D, 4N-R; Figure 5G-no stats indicated in figure legend.*

We sincerely thank the reviewer for highlighting the inappropriate use of statistical tests in our original submission. We have re-analyzed all data using the ANOVA-based methods as suggested in the specific detail. We confirm that these changes do not alter the overall interpretation of our results but provide a more robust and statistically sound foundation for our conclusions. We changed "Differences were determined by two-tailed independent Student's t-test" to "Statistical significance was determined by one-way ANOVA followed by Tukey's test (C, Q-U, different letters above the error bar indicate statistical differences at  $P < 0.05$ ) or Student's t-test (D) (\*,  $P < 0.05$ ; \*\*,  $P < 0.01$ ; NS, no significant difference)."



In lines 719-721 we added “Statistical significance was determined by one-way ANOVA followed by Tukey's test (E, different letters above the error bar indicate statistical differences at  $P < 0.05$ ) or Student's  $t$ -test (B, H) (\*,  $P < 0.05$ ; \*\*,  $P < 0.01$ ; NS, no significant difference).” in line 745-747.

**Reviewer #1 (Significance):**

*The data presented in this manuscript provides important context towards the connection between the DHH pathway, Sf1, and steroidogenesis.*

*The audience would likely include developmental biologists, including those related to differentiation of any hormone producing cell type and especially those focused on steroidogenesis onset. Clinical interests will be related to sex determination and differentiation, especially related to male sex phenotype differentiation. Basic scientists will be especially interested.*

*Expertise: mouse fetal testis differentiation and maturation, steroidogenesis, hedgehog, sf1. Good fit except for the animal model, but they are surprisingly similar.*

**Reviewer #2 (Evidence, reproducibility and clarity):**

*In this work, Zhao et al., investigated the role of Dhh signaling pathway in the proliferation and differentiation of leydig lineage cells in the testes of Nile tilapia, an economic important farmed fish. By generating dhh mutants, the authors showed that loss of Dhh in tilapia recapitulated mammalian phenotypes, characterized by testicular hypoplasia and androgen insufficiency. A previous established TSL line was used to rescue the deficits in dhh<sup>-/-</sup> testes, which demonstrated that Dhh regulates the differentiation of SLCs rather than their survival. By generating mutant TSL lines, the authors aimed to identify the downstream players under Dhh in tilapia. Based on the data, the authors propose that a dhh-ptch2-gli1-sf1 axis exists in leydig cell lineage development.*

*How secreted dhh from Sertoli cells affect the Leydig cells remains elusive. While previous studies have revealed the paracrine role of Sertoli cell secreted Dhh in the regulation of Leydig cell development and maturation, the authors provided some new insights into the issue using tilapia as a model. Unfortunately, this work is not well performed, and the conclusions are not well supported by the current data. And to reach logic conclusions, more meaningful experiments should be performed, and more convincing data should be provided.*

**Strength:**

*The authors used genetic mutants, TSL lines, and cell transplantation techniques to address the questions. The manuscript is technically sound, and overall is well-written.*

**Limitations:**

*Experimental design should be optimized, and more convincing data should be provided to reach solid conclusion.*

*(1) The SLCs (stem leydig cells) used in this work. The SLC line was established from 3-month-old immature XY tilapia. The authors claimed that this line is a SLC line only because they express a few Leydig markers such as pdgfra and nestin. However, in my opinion, the identity of the cell line is not clear. It is suggested to perform more experiments, including flow cytometry assay or single cell RNA sequencing analysis, to further characterize this line, to demonstrate that this line is a real SLCs that are*

*equivalent to the SLCs in 3-month testes of tilapia. According to the previous publication (2020), the information about the line was not well presented.*

We thank the reviewer for this comment regarding the characterization of the TSL cell line. The identity of TSL as a stem Leydig cell line was rigorously established in our previous publication (Huang *et al.*, 2020), which provided comprehensive molecular, *in vitro*, and *in vivo* functional evidence that meets the definitive criteria for an SLC. This includes its stable expression of established SLC markers (*pdgfra*, *nestin*, *coup-tfii*), its capacity to differentiate into steroidogenic cells producing 11-KT *in vitro*, and most critically, its ability to colonize the testicular interstitium, differentiate into Leydig cells, and restore androgen production upon transplantation *in vivo*.

In direct response to the reviewer's point, we have revised the Introduction of our manuscript to provide a more detailed and clear description of the TSL line's origin and validation (lines 95-105) as “Furthermore, a stem Leydig cell line (TSL) has been established from the testis of a 3-month-old Nile tilapia. TSL expresses platelet-derived growth factor receptor  $\alpha$  (*pdgfra*), *nestin*, and chicken ovalbumin upstream promoter transcription factor II (*coup-fla*), which are usually considered as SLC-related markers in several other species. Notably, this cell line exhibits the capacity to differentiate into 11-ketotestosterone (11-KT)-producing Leydig cells both *in vitro* and *in vivo*. When cultured in a defined induction medium, TSL cells differentiate into a steroidogenic phenotype, expressing key steroidogenic genes including *star1*, *star2*, and *cyp11c1*, and producing 11-KT; upon transplantation into recipient testes, TSL cells successfully colonize the interstitial compartment, activate the expression of steroidogenic genes, and restore 11-KT production”, ensuring that readers can fully appreciate its well-founded identity as a SLC model without needing to consult the original publication. We are confident that the existing body of evidence solidly supports all conclusions drawn from its use in this study.

*(2) How loss of dhh affects testicular and the leydig cell lineage development are not clearly investigated. In the current manuscript, the characterization of dhh mutant was not enough and lack of in-depth investigation. The authors primarily looked at testes at 90 dph when Leydig cell lineage was well developed. In my opinion, this time was too late. To investigate the earlier events that are affected by loss of dhh, I suggested to perform experiments at earlier time points, in particular around the initiation stages of the sex differentiation and Leydig cell specification/maturation.*

We thank the reviewer for this insightful comment. We agree that a thorough developmental analysis is crucial. In response to this point, we have now performed an in-depth investigation at earlier stages to precisely define the phenotype onset.

Our revised manuscript includes new data from a developmental time-course analysis. While our initial characterization included 5, 10, and 20 dah, we now identified 30 dah as the critical window for Leydig cell differentiation onset, which was also supported by prior work (Zheng *et al.*). Our new immunofluorescence data at 30 dah now clearly show that Cyp11c1-positive cells are present in wild-type testes but are entirely absent in *dhh*<sup>-/-</sup> mutants (Fig. S7). This finding pinpoints the initial failure of SLC differentiation.

We have integrated this key finding into the Discussion (lines 234-239) as “To define the onset of Leydig cell differentiation, we performed a developmental time-course analysis. This revealed that Cyp11c1-positive steroidogenic cells first appear in wild-type testes at 30 dah, while being conspicuously absent in *dhh*<sup>-/-</sup> mutants at this same stage (Fig. S7). This clear temporal pattern establishes ~30 dah as the developmental window when SLCs initiate their differentiation program in the Nile tilapia.”

Concurrently, our analysis of the 90 dah timepoint remains vital, as it represents a mature stage with robust spermatogenesis and a stabilized somatic niche. This allows for a

comprehensive assessment of the ultimate functional consequences of the early differentiation block, including its impact on germ cell support and overall testicular architecture.

Thus, our study now provides a complete developmental perspective: the 30 dah timepoint identifies the initiation of the Dhh-dependent defect, while the 90-dah analysis reveals the mature, functional outcomes within the intact testicular niche.

*(3) The authors claimed that there was a *ptch2-gli1-sf1* axis. The conclusion was drawn largely based on data that generated from the *in vitro* cultured TSL line. More data from genetic mutant tilapia are required to support the conclusion.*

We thank the reviewer's insightful comments regarding the need for robust *in vivo* validation. In fact, our conclusion of a Dhh-Ptch2-Gli1-Sf1 axis is supported by an integrated experimental strategy, combining key *in vivo* evidence with targeted *in vitro* analyses to build a coherent model.

(1) Evidence for Ptch2 as the key receptor: The role of Ptch2 is supported by a pivotal *in vivo* genetic experiment. The observation that the *dhh*<sup>-/-</sup> testicular phenotype is fully rescued in *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> double mutants provides compelling genetic evidence that Ptch2 is the essential receptor for Dhh *in vivo* (Fig. 4E-U). We acknowledge that the early embryonic lethality of global *ptch1* mutation precludes its functional analysis in postnatal testis development. Therefore, while our data strongly nominate Ptch2 as the principal receptor, we have qualified our conclusions in the revised manuscript to reflect that a role for Ptch1 cannot be definitively excluded without Leydig cell-specific conditional knockout models.

(2) Evidence for Gli1 and its regulation of Sf1: The role of Gli1 as the key transcriptional effector was efficiently identified using our well-characterized TSL system, a valid approach for dissecting this highly conserved signaling cascade. The functional connection between Gli1 and Sf1 is supported by multiple lines of evidence: transcriptomic profiling, promoter analysis, luciferase reporter assays (including a new cold probe competition experiment), and most importantly, *in vivo* functional validation via SLC transplantation. The latter demonstrated that Sf1 is both necessary and sufficient for SLC differentiation within the testicular niche (Fig. 5).

In direct response to the reviewer's points, we have thoroughly revised the manuscript text to ensure all claims are accurately stated, particularly regarding the receptor specificity and the nature of the Gli1-Sf1 regulatory relationship. We believe our study provides a solid foundation for the proposed signaling axis.

*Overall, better experimental design should be planned, including the rescue experiments. Some key information was missed. For instance, the identity of the stem Leydig cells was not clearly presented.*

We have explained it in point #1.

*Figures:*

*Figure 1: The authors described the phenotypes at 90 dph. Loss of *dhh* led to severe phenotypes in testicular formation, as evidenced by defective formation of Vasa, a germline stem cell marker; loss of expression of *cyp11c1*, a Leydig cell marker; and loss of *sycp3*, a marker of meiosis of spermatogonia.*

*However, in my opinion, 90 dph was too late. To investigate the role of *dhh* in Leydig cell lineage, the authors are suggested to focus on earlier developmental stages when the sex differentiation and maturation of Leydig cells occur. This work is actually a development biology one that investigates how *dhh* loss in Sertoli cells affects the development of*

*Leydig cells. The careful characterization of earliest testicular phenotypes of dhh mutant is very important.*

We have explained it in point #2.

*Figure 2: Please clarify the logic for performing rescue experiments using 11-KT. Provided the critical role of 11-KT in the testis development and spermatogenesis, it was not unexpected that 11-KT treatment can rescue most of the cell types in testes. If dhh is absolutely required for LC lineage development maturation, adding 11-KT at 30 dph will not have an effect. Why not perform rescue experiments using Dhh protein?*

We thank the reviewer for this insightful comment, which allows us to clarify the logical progression of our experimental design, a process central to genetic discovery.

When we first characterized the *dhh*<sup>-/-</sup> mutant, we observed a complex suite of phenotypes: testicular hypoplasia, arrested germ cell development, a profound deficiency of Leydig cells, and drastically low androgen levels. A primary challenge was to distinguish which defects were direct consequences of losing Dhh signaling and which were secondary effects of the overall testicular failure.

We therefore employed a classic genetic strategy: phenotypic dissection through targeted rescue. The 11-KT rescue experiment was designed to test a foundational hypothesis: Are the severe testicular defects in *dhh*<sup>-/-</sup> mutants primarily a consequence of the systemic androgen deficiency? The results provided a pivotal and clear answer: while 11-KT treatment partially rescued germ cell development and testicular structure, it completely failed to restore the population of Cyp11c1-positive Leydig cells. This critical finding allowed us to dissociate the phenotypes, demonstrating that the Leydig cell defect is a primary, cell-autonomous consequence of Dhh loss, not a secondary effect of low androgen.

This conclusion logically propelled the next phase of our research: to shift focus from systemic hormone action to the local, niche role of Dhh in regulating the Leydig lineage directly. This led directly to the TSL transplantation experiments and the mechanistic dissection of the Ptch2-Gli1-Sf1 axis within SLCs.

Regarding the use of Dhh protein, we agree it is a complementary approach. However, producing biologically active, recombinant Hedgehog ligand is challenging due to its essential dual lipid modification, which is required for solubility and activity. Our transplantation experiments with TSL-*OnDhh* cells (Fig. 3) functionally demonstrate that providing Dhh signaling in a cell-autonomous manner is sufficient to rescue differentiation, thereby directly addressing the core question without the need for recombinant protein.

*Figure 3. The authors showed that in dhh-/- testes, TSL engrafted equivalently but failed to express Cyp11c1. This result was strange which raised a question about the identity of the TSLs, as I have mentioned above. The authors claimed that the TSLs are stem Leydig cells, which I doubt. Additional data should provided to support the statement.*

*In the testicular environment, the transplanted TSLs should be able to colonize and differentiate into more mature leydig cells. Only a small portion of the PKH26-labeled TSLs became Cyp11c1 positive after transplantation, can the authors comment this observation?*

*To address "Mutation of dhh blocks SLC differentiation", the authors should first carefully examine the TSL lineage development using dhh mutant. Then, investigate how loss of dhh disrupts the cross talk between Sertoli cells and Leydig cells. why bother performing transplanted TSLs? Please clarify. Why not perform rescue experiments using Dhh protein at appropriate developmental stages?*

We thank the reviewer for these comments, which allow us to clarify the rationale and interpretation of our key experiments.

(1) We have provided comprehensive evidence establishing the TSL line as a SLC line (Response to Point #1). The observation that WT TSL cells engraft but fail to differentiate in the *dhh*<sup>-/-</sup> testicular environment is not strange; it is, in fact, the core and most crucial finding of this experiment. It provides direct functional evidence that the *dhh*<sup>-/-</sup> niche lacks the essential signals required to initiate SLC differentiation, consistent with the severe deficiency of endogenous Cyp11c1<sup>+</sup> cells in these mutants (Fig. 1I-J', N).

(2) The reviewer's concern about "only a small portion" of cells differentiating is based on a misunderstanding. Our quantitative data (Fig. 3F) show that approximately 78% of the transplanted PKH26<sup>+</sup> TSL cells successfully differentiated into Cyp11c1<sup>+</sup> cells in WT hosts. This high efficiency robustly demonstrates the differentiation potential of TSL cells and the permissiveness of the WT niche. The near-zero differentiation rate in the *dhh*<sup>-/-</sup> host (Fig. 3F) starkly highlights the specific and severe defect in the mutant microenvironment.

(3) The TSL transplantation experiment was the most direct strategy to test why Cyp11c1<sup>+</sup> cells are absent in *dhh*<sup>-/-</sup> testes. It allowed us to distinguish between a failure in SLC differentiation and other possibilities (e.g., cell death). The finding that functional SLCs cannot differentiate in the mutant niche logically directed our subsequent focus onto the cell-intrinsic molecular mechanism (the Ptch2-Gli1-Sf1 axis) within the Leydig lineage. While Sertoli-Leydig crosstalk is an important area, it was beyond the scope of this study aimed at defining the intrinsic differentiation pathway.

(4) Regarding Dhh protein rescue, generating bioactive, lipid-modified recombinant Hh protein is technically challenging. Our transplantation of TSL-*OnDhh* cells (Fig. 3) functionally demonstrates that providing Dhh signaling in a cell-autonomous manner is sufficient to rescue differentiation, effectively addressing this question without the need for recombinant protein.

*Figure S3. "To assess whether dhh mutation affects androgen-producing cells outside Leydig cells, 11-KT levels were analyzed during early testicular development before SLCs differentiation. IF analyses revealed that no Cyp11c1 positive cells were present in the testes of XY WT fish at 5, 10, and 20 dah, indicating that SLCs had not yet differentiated at these stages (Fig. S3A-C). Tissue fluid 11-KT levels showed no significant differences between WT and dhh-/- XY fish at 5, 10, and 20 dah (Fig. S3D)". These observations suggested that loss of dhh does not affect the specification of SLCs, but affect its differentiation into mature LCs. The differentiation of Cyp11c1 should be later than 20 dah. So when is the earliest time point for formation of Cyp11c1 positive cells, and how loss of dhh affect this? These are important questions to answer.*

We agree with the reviewer's interpretation that our data suggest *dhh* loss affects SLC differentiation rather than initial specification. In direct response to the need for earlier timepoints, we have now performed and included an analysis at 30 dah, which we identified as the critical window for Leydig cell differentiation onset. Our new data (Fig. S7) show that Cyp11c1<sup>+</sup> cells are present in WT testes but are entirely absent in *dhh*<sup>-/-</sup> mutants at this stage. This precisely pinpoints the initiation of the phenotypic divergence and establishes ~30 dah as the developmental window when Dhh signaling is required to drive SLC differentiation. Our study therefore now provides a complete developmental perspective, from the initial failure at 30 dah to the mature functional outcomes at 90 dah.

*Figure 4. The authors generated ptch1/2 mutant TSL lines, and luciferase assay was performed, and based on the results, the authors concluded that Ptch2, but not Ptch1, is specifically required for transducing Dhh signals in TSLs. The conclusion was only based*



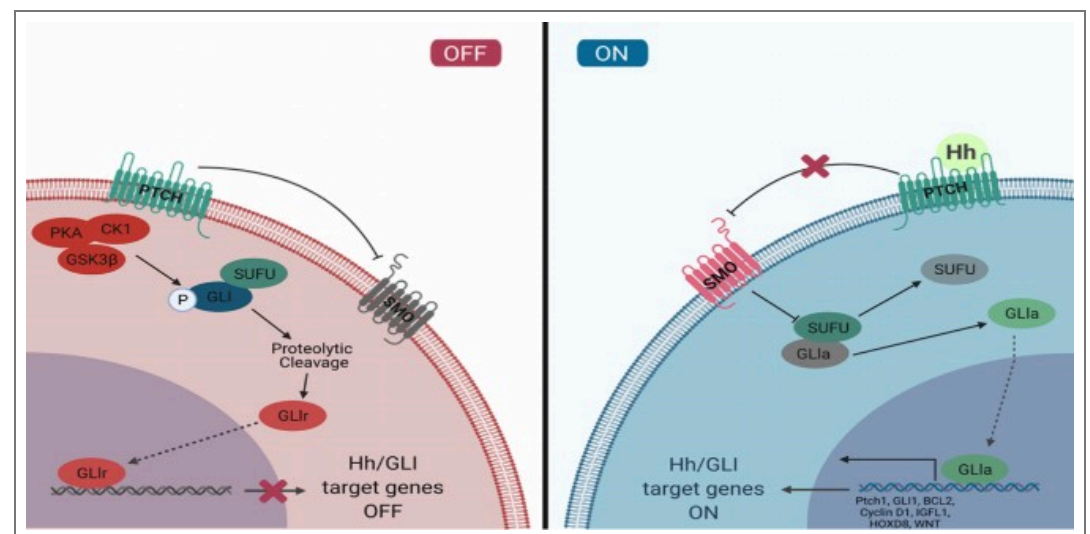
on luciferase assay using TSLs. Whether this was the case in testes at animal level is not clear. Clearly, more genetic experiments, using *ptch* mutants, should be performed to substantiate this.

The authors stated “*Ptch2* acts as the obligate receptor for *Dhh* signaling during testis development”. If *ptch2* is required for TSL lineage, why *ptch2*<sup>-/-</sup> testes exhibited no significant differences in testicular histology and Leydig cell (*Cyp11c1*+) populations and serum 11-KT levels? This contradictory statement needs to be addressed.

We thank the reviewer for these critical comments, which allow us to clarify the logic underlying our conclusions regarding *Ptch2*.

(1) *In Vivo* Genetic Evidence for *Ptch2*: Our conclusion that *Ptch2* is the primary receptor for *Dhh* is not based solely on the TSL luciferase assays. It is definitively supported by a key *in vivo* genetic experiment: the complete phenotypic rescue in the *dhh*<sup>-/-</sup>;*ptch2*<sup>-/-</sup> double mutants (Fig. 4F-R). In genetic terms, the loss of the receptor (*ptch2*) suppressing the phenotype caused by the loss of the ligand (*dhh*) is classic evidence for a ligand-receptor relationship within a linear pathway. This *in vivo* evidence strongly substantiates *Ptch2*'s role at the animal level. The early embryonic lethality of *ptch1* mutants precludes a similar *in vivo* test for *Ptch1* in postnatal testis development.

(2) Addressing the Apparent Contradiction of the *ptch2*<sup>-/-</sup> Phenotype: The reviewer raises an excellent point, which stems from the fundamental biology of the Hh pathway as shown in Author response image 1. *Ptch* receptors are inhibitory. In the absence of ligand, *Ptch* suppresses pathway activity.



**Author response image 1. The canonical Hh signaling pathway.** In the *dhh*<sup>-/-</sup> mutant, the pathway is suppressed due to unopposed *Ptch* activity, leading to a failure in SLC differentiation. In the *ptch2*<sup>-/-</sup> mutant, this key inhibitory brake is removed, leading to constitutive activation of the pathway. The fact that *ptch2*<sup>-/-</sup> testes are normally indicates that this level of pathway activation is not detrimental and, crucially, is sufficient to support wild-type levels of Leydig cell development and steroidogenesis. This lack of a phenotype in the receptor mutant, contrasted with the severe ligand mutant phenotype, is a common and expected observation in signaling pathways where the receptor acts as a tonic inhibitor.

In summary, the normal development of *ptch2*<sup>-/-</sup> testes is not contradictory but is entirely consistent with its role as the inhibitory receptor for *Dhh*. The severe phenotype in *dhh*<sup>-/-</sup> mutants and its specific rescue by removing *ptch2* provides compelling genetic evidence for

their functional relationship. We have revised the text throughout the manuscript to ensure these conclusions are accurately stated.

*Figure 5. The authors generated gli1/2/3 mutant TSL lines, and luciferase assay was performed, and based on the results, the authors concluded that Gli1, but not Gli2/3, was specifically required for transducing Dhh signals in TSL cells. The conclusion is drawn, only based on luciferase assay using TSLs. Whether this was the case in testes at animal level is not clear. Clearly, more genetic experiments should be performed to substantiate this, using the gli mutant fish.*

*To identify Gli1-dependent targets in SLCs, the authors compared transcriptomes of TSLWT, Dhh-overexpressing (TSL-OnDhh), Gli1-overexpressing (TSL-OnGli1), and SAG-treated (TSL+ SAG) TSL cells. While this experiment can be used to identify dhh target genes, it is better to use gli mutant cell lines. Since the authors have generated gli1/2/3 mutants, why not using these mutant fish to identify/confirm the Gli targets?*

We thank the reviewer for these comments.

(1) We acknowledge that Gli1 as the key transcriptional effector is primarily based on our in vitro evidence using the TSL cell line. We have revised the manuscript accordingly to ensure this is stated precisely, avoiding overstatement.

(2) Concerning the transcriptomic analysis, the reviewer suggests using *glis* mutant cell lines. While this is a valid approach, our strategy of profiling pathway activation (via Dhh/Gli1 overexpression or SAG treatment) was deliberately chosen to provide a high signal-to-noise ratio for identifying genes that are positively upregulated during the differentiation process. Analyzing loss-of-function mutants under basal conditions can be confounded by potential compensatory mechanisms among the Gli family members, potentially masking the specific transcriptional signature of pathway activation we sought to capture.

By the way, we have generated *gli1/2/3* mutant TSL cell lines for the functional luciferase assays, but we have not generated the corresponding *glis* mutant fish lines, which would represent a substantial new line of investigation.

#### **Reviewer #2 (Significance):**

*While previous studies have revealed the paracrine role of Sertoli cell secreted Dhh in the regulation of Leydig cell development and maturation, the authors provided some new insights into the issue using tilapia as a model.*

#### **Reviewer #3 (Evidence, reproducibility and clarity):**

##### **Summary**

*The authors investigate the Dhh signaling pathway in Leydig cell differentiation in the tilapia model. They generated multiple mutant lines in different hedgehog pathway components and utilized a Leydig stem cell line to interrogate Leydig cell differentiation. Through this analysis, the authors demonstrate that Dhh regulates Leydig differentiation rather than survival. They also found that Ptch2 is the specific receptor that mediates signaling to promote Leydig differentiation and that Gli1 is the primary Gli involved. Furthermore, they show that a known regulator of Leydig cell development and function, SF1, is a downstream transcriptional target. Overall, the study identifies previously unknown information as to how Dhh signaling regulates Leydig cell development, which is necessary for testosterone production by the testis.*

##### **Major Comments**

*(1) In the RNAseq analysis is not clear exactly how the 33 "up-regulated" genes were identified. What was the methodology for identification of these genes? Some of the genes were down-regulated or not different in the OnGli condition and some in the OnDhh condition were not differentially expressed, as shown in Fig S8B. Therefore, it is unclear why all 33 genes are classified as upregulated "across all three conditions".*

We have clarified this methodology in the Materials and Methods section in line 452-454: "Differentially expressed genes (DEGs) were identified for each condition (TSL-*OnDhh*, TSL-*OnGli1*, TSL+SAG) compared to TSL-WT controls using edgeR (threshold: FDR < 0.05,  $|\log_2(\text{foldchange})| \geq 1.5$ ). And we Added relevant information in the Results section in line 198-202: we performed transcriptomic profiling of TSL cells under conditions of pathway activation: Dhh overexpression (TSL-*OnDhh*), Gli1 overexpression (TSL-*OnGli1*), and SAG treatment (TSL+SAG). Comparative RNA-seq analysis identified a core set of 33 genes consistently upregulated across all three conditions (Fig. 5C, S6A)."

We have also updated Fig. S8B to include a clear value and to better visualize the FPKM value levels of these 33 genes across the conditions.

*(2) In figure 4A (and possibly B), it appears that ptch RNA is in the nucleus of the cell. Why would the RNA be primarily in the nucleus? Is the RNA detection accurate? Were controls done? The methods state that sense probes were made but no how they compared to the antisense probes. This comment can also be applied to the gli FISH, particularly gli3 (Figure 5).*

This is an excellent observation. We speculate that the apparent nuclear signal may be due to strong transcriptional activity in the nucleus. To confirm the specificity of our FISH experiment, we performed FISH with sense RNA probes as negative controls for all genes (*ptch1*, *ptch2*, *gli1*, *gli2*, *gli3*), and no specific signals were observed (see New Fig. S9).

#### Minor comments

*(1) In the introduction, please include information as to when tilapia reach sexual maturity*

We have added this information to the Introduction in line 91-92: early sexual maturity (approximately 3 months after hatching for males and 6 months after hatching for females).

*(2) When first mentioning experiments that use the PKH26 dye, please give a brief description of the dye in the text of the results. This is described in the methods but it would be helpful to have some information about what PKH26 is in the results to more easily understand the figure and experimental design.*

We have added a brief description in the Results section in line 151-152: "To dissect Leydig cell lineage impairment in *dhh*<sup>-/-</sup> testes, we transplanted the TSL labeled with PKH26 (a fluorescent red hydrophobic membrane dye that enables tracking of transplanted cells) into WT and *dhh*<sup>-/-</sup> testes (Fig. 3A)."

*(3) In the statistical analysis section of the methods, the authors state that two-tailed t-tests were performed however in the figure legends it states that ANOVA was done for some of the statistical analysis. Please clarify this.*

We have updated the Statistical Analyses section in Methods to clarify in line 472-476: "A two-tailed independent Student's *t*-test was used to determine the differences between the two groups. One-way ANOVA, followed by Tukey multiple comparison, was used to determine the significance of differences in more than two groups. *P* < 0.05 was used as a threshold for statistically significant differences."

(4) Figures - in figures that have charts with the Y-axis labeled as "relative positive cells", or similar, please explain what exactly is meant by "relative". What is it relative to?

We have revised all relevant Y-axis labels and figure legends to explicitly state the quantification method. For example, we now use: "Vasa<sup>+</sup> / DAPI<sup>+</sup> (%)", "Sycp3<sup>+</sup> / DAPI<sup>+</sup> (%)" or "Cyp11c1<sup>+</sup> / DAPI<sup>+</sup> (%)".

(5) Figure 1: please point out the testes in panels A and B

We have indicated the position of the testes with arrows in Figures 1A and B.

(6) In figure 4, it would be helpful for the WT images from S7 moved to fig 4.

We have moved representative WT images from Fig. S7 into Fig. 4 for easier comparison with the mutant phenotypes.

(7) Figure 4E: Are the yellow bars comparable to each other. Is there any significance to the increased luciferase with 8xGli in *ptch2*<sup>-/-</sup> as compared to the other genotypes?

We thank the reviewer for this astute observation. Yes, the yellow bars are directly comparable, and the elevated basal luciferase activity of the 8xGli reporter in the *ptch2*<sup>-/-</sup> TSL cells is indeed significant and expected. The genetic ablation of *ptch2* removes this inhibition, leading to ligand-independent, constitutive activation of the downstream signaling cascade. The observed increase in basal reporter activity in the *ptch2*<sup>-/-</sup> cells is a classic manifestation of this mechanism.

The primary objective of this experiment was to test the cells' responsiveness to Dhh stimulation across genotypes. The key finding is that while wild-type and *ptch1*<sup>-/-</sup> cells showed a significant response to Dhh, the *ptch2*<sup>-/-</sup> cells-which already exhibited high basal activity-were completely unresponsive. This combination of constitutive activation and ligand insensitivity in the *ptch2*<sup>-/-</sup> genotype provides particularly strong genetic evidence that Ptch2 is the essential receptor mediating Dhh signal transduction in this system.

(8) Figure 5G: please include what exactly what each construct name stands for in the figure legend

We have expanded the legend for Fig. 5G to define each construct.

(9) Figure S8B: please include what the values in the table are (eg are these the significance values?)

We have updated the caption for Figure S8B (now Figure S6B): "The FPKM value for each gene in each sample is indicated within the squares. The color gradient from blue to red reflects low to high expression levels per row (gene)."

### Reviewer #3 (Significance):

#### Strengths and limitations:

*The genetics of the tilapia system and the availability of the tilapia Leydig stem cell lines were particular strengths of this study. The study utilizes fish genetics to genetically interrogate the Dhh signaling pathway in Leydig cell development through generation and analysis of mutant lines. The tilapia Leydig stem cell line was an integral part of this study as it allowed for genetic and chemical manipulation of Dhh signaling in undifferentiated Leydig cells and, through transplantation into testes, allowed for analysis of how Leydig cell differentiation was affected.*

#### Advance:

*The study makes significant advances as to how Dhh signaling instructs Leydig cell differentiation, including identification of the Ptch receptor and Gli transcription factor that function downstream of Dhh in this process. Furthermore, they identify a direct link between Dhh signaling and Sf1 expression, which is known to be important for Leydig cell function.*

*Audience:*

*This study will be of particular interest to reproductive biologists, endocrinologists, and developmental biologists. The study may also be of interest to researchers and physicians investigating cancers that are promoted by androgens produced by Leydig cells of the testis.*

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